Series & Progressions

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Series & Progressions

Introduction :

E.g.
0, 7, 26, ...(rule is n^3-1)
1, 4, 7, 10, ...
2, 4, 6, 8, ... etc.

Note :
Minimum number of terms in a sequence should be 3.

Definition

A sequence is a set of terms which may be algebraic, real or complex numbers, written according to definite rule and the series thus formed is called a **progression**.

Arithmetic Progression :
It is a sequence whose terms increase or decrease by a fixed number. Fixed number is called the common difference. If 'a' is the first term and 'd' is the common difference, then the standard appearance of an A. P. is

$$a,(a+d),(a+2d),...,\left(a+\overline{n-1}d\right)$$

n^{th} term = $T_n = a+(n-1)d$

Note :

If $d > 0 \Rightarrow$ increasing A. P.
If $d < 0 \Rightarrow$ decreasing A. P.
If $d = 0 \Rightarrow$ all terms remain same

Q.1 If 6th and 11th term of an A.P. are respectively 17 and 32. Find the 20th term.

Sol. $T_6 = 17 \Rightarrow a+5d = 17$ ...(i)
$T_{11} = 32 \Rightarrow a+10d = 32$ ...(ii)
(ii) – (i)
$5d = 15 \Rightarrow d = 3$
Putting in (1)
$a + 5(3) = 17 \Rightarrow a = 2$
now, $T_{20} = a + 19d$
$=2 + 19(3) = 59$

Q.2 In an A. P. if $t_p = q$ and $t_q = p$, then find the rth term.

Sol. $t_p = q \Rightarrow a + (p - 1)d = q$ ...(i)
$t_q = p \Rightarrow a + (q - 1)d = p$ ...(ii)
(i) – (ii)
$q - p = (p - q)d$

$d = -1$
from (1), $a + (p - 1)(-1) = q \Rightarrow a = p + q - 1$
Now
$t_r = a + (r - 1)d = (p + q - 1) + (r - 1)(-1)$
$= p + q - r$

Q.3 In an A. P. if $a_2 + a_5 - a_3 = 10$ and $a_2 + a_9 = 17$, then find the 1st term and the common difference

Sol. $a_2 + a_5 - a_3 = 10$
$(a + d) + (a + 4d) - (a + 2d) = 10$
$a + 3d = 10$ ...(i)
$a_2 + a_9 = 17$
$(a + d) + (a + 8d) = 17$
$2a + 9d = 17$ ...(ii)
(ii) $- 2 \times$ (i)
$3d = -3 \Rightarrow d = -1, a = 10 - 3d = 13$

Q.4 If p^{th}, q^{th} & r^{th} term of AP are respectively a, b and c, then prove that
$a(q - r) + b(r - p) + c(p - q) = 0$

Sol. $T_p \Rightarrow A + (p - 1) D = a$ ...(i)
$T_q \Rightarrow A + (q - 1) D = b$ ...(ii)
$T_r \Rightarrow A + (r - 1) D = c$ ...(iii)
(i) $-$ (ii) $\Rightarrow (p - q) D = a - b$
(ii) $-$ (iii) $\Rightarrow (q - r) D = b - c$
(iii) $-$ (i) $\Rightarrow (r - p) D = c - a$
$a(q - r) + b(r - p) + c(p - q)$

$$= \frac{a(b-c)}{D} + \frac{b(c-a)}{D} + \frac{c(a-b)}{D}$$

$$= \frac{1}{D}\{ab-ac+bc-ab+ac-bc\} = 0$$

Hence proved.

Q.5 If a, b, c, d are distinct integers in an A.P. such that $d = a^2 + b^2 + c^2$, then find the value of $a + b + c + d$.

Sol. Let $b = a + \alpha$, $c = a + 2\alpha$, $d = a + 3\alpha$
now $d = a^2 + b^2 + c^2$
$a + 3\alpha = a^2 + (a + \alpha)^2 + (a + 2\alpha)^2$
$a + 3\alpha = 3a^2 + 6a\alpha + 5\alpha^2$
$\Rightarrow 5\alpha^2 + 3(2a - 1)\alpha + a(3a - 1) = 0$ ...(i)
now $\alpha \in R \Rightarrow D \geq 0$
$\Rightarrow 9(2a - 1)^2 - 4.5.a(3a - 1) \geq 0$
$\Rightarrow 36a^2 - 36a + 9 - 60a^2 + 20a \geq 0$
$\Rightarrow 24a^2 + 16a - 9 \leq 0$

$\Rightarrow a \in \left[\frac{-4-\sqrt{70}}{12}, \frac{-4+\sqrt{70}}{12}\right]$

$\because a \in I$ hence $a = 0$ or -1

but for $a = 0$, $\alpha = 0$ or $\frac{3}{5}$ which is not permissible

hence $a = -1 \Rightarrow \alpha = 1, \frac{4}{5}$ (Rejected)

$a = -1, b = 0, c = 1, d = 2$

$\Rightarrow a + b + c + d = 2$

Sum of n terms of an A. P. :

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_n = \frac{n}{2}(a + \ell) \text{ where } \ell = a + (n-1)d$$

HIGHLIGHTS ABOUT AN A.P.

(i) If each term of an A.P. is increased, decreased, multiplied or divided by the same non zero number, then the resulting sequence is also an AP.

(ii) Three numbers in AP can be taken as a – d, a, a + d; four numbers in AP can be taken as a - 3d, a - d, a + d, a + 3d; five numbers in AP can be taken as a-2d, a-d, a, a + d, a + 2d & six terms in AP can be taken as a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d etc.

(iii) The common difference can be zero, positive or negative.

(iv) The sum of the two terms in an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms.

(v) If the number of terms in an A.P. is even then take it as 2n and if odd then take it as (2n+1)

(vi) For any series $T_n = S_n - S_{n-1}$, for $n > 1$.

(vii) If a, b, c are in A.P. $\Rightarrow 2b = a + c$

Points to Remember!!!

(i) Sum of first n natural numbers is $\frac{n(n+1)}{2}$

(ii) Sum of first n odd natural numbers is n^2

(iii) Sum of first n even natural is n(n + 1)

Q.1 The first term of an AP is 5, the last is 45, and the sum 400, find the number of terms and the common difference.

Sol. Given $a = 5$

$\ell = a + (n - 1)d = 45$ ...(1)

Also, $S_n = \frac{n}{2}[a + \ell] \Rightarrow \frac{n}{2}(50) = 400 \Rightarrow n = 16$

number of terms = 16

from (1)

$5 + (16 - 1)d = 45 \Rightarrow d = \frac{40}{15} = \frac{8}{3}$

Q.2 The sum of first 3 terms of an AP is 27 and the sum of their squares is 293 then find S_n

Sol. $a + (a + d) + (a + 2d) = 27$

$\Rightarrow 3(a + d) = 27 \Rightarrow a + d = 9$ ...(1)

now, $a^2 + (a + d)^2 + (a + 2d)^2 = 293$

$(9-d)^2 + 9^2 + (9+d)^2 = 293$

$243 + 2d^2 = 293 \Rightarrow 2d^2=50$

$d^2 = 25 \Rightarrow d = \pm 5$

from (1)

if $d = 5 \Rightarrow a = 4$

if $d = -5 \Rightarrow a = 14$

Now, for $a = 4, d = 5 \Rightarrow S_n = \frac{n}{2}[8 + (n-1)5] = \frac{n}{2}(3 + 5n)$

For $a = 14, d = -5 \Rightarrow S_n = \frac{n}{2}[28 + (n-1)(-5)] = \frac{n}{2}(33 - 5n)$

Q.3 In an AP of which 'a' is the 1st term, if the sum of the 1st p terms is equal to zero, show that the sum of the next q terms is $-\left(\frac{aq(p+q)}{p-1}\right)$.

Sol. Given $S_p = 0$

$\frac{p}{2}[2a + (p-1)d] = 0 \Rightarrow 2a + (p-1)d = 0$

$d = \frac{-2a}{p-1}$ ...(1)

now sum of next q terms

$= T_{p+1} + T_{p+2} + ...$(q terms)

$= \frac{q}{2}[2(a + pd) + (q-1)d]$

$$= \frac{q}{2}\left[2a + (2p + q - 1)\left(\frac{-2a}{p-1}\right)\right]$$

$$= \frac{q}{2}\left[\frac{2ap - 2a - 4ap - 2aq + 2a}{(p-1)}\right]$$

$$= \frac{q}{2}\left[\frac{-2a(p+q)}{p-1}\right] = -aq\left(\frac{p+q}{p-1}\right)$$

Q.4 Find the value of the sum $\sum_{k=0}^{359} k \cos k°$

Sol. Sum

$$S = 0\cos0° + 1\cos1° + 2\cos2° + 3\cos3° + \ldots + 359\cos359°$$

$\because \cos359° = \cos1°$

$\cos358° = \cos2°$

$\vdots$

$\cos181° = \cos179°$

hence

$$S = 360\{\cos1° + \cos2° + \ldots + \cos179°\} + (\cos180°)180$$

$n=179,\ \alpha = 1,\ \beta = 1$

$$S = 360 \times \frac{\sin\left(\frac{179°}{2}\right)}{\sin\left(\frac{1°}{2}\right)}\cos(90°) - 180$$

$$S = -180$$

Q.5 In an AP if $T_m = 1/n$ and $T_n = 1/m$ then show that $S_{mn} = \frac{1}{2}(mn+1)$

Sol. $T_m = a + (m-1)d = \frac{1}{n}$ ...(1)

$T_n = a + (n-1)d = \frac{1}{m}$ ...(2)

(1) – (2)

$$(m-n)d = \frac{m-n}{mn} \Rightarrow d = \frac{1}{mn}$$

from (1)

$$a + (m-1)\frac{1}{mn} = \frac{1}{n} \Rightarrow a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{n}$$

$$\Rightarrow a = \frac{1}{mn}$$

now $S_{mn} = \frac{mn}{2}\{2a + (mn-1)d\}$

$$= \frac{mn}{2}\left\{\frac{2}{mn} + \frac{mn-1}{mn}\right\} = \frac{1}{2}(mn+1)$$

Q.6 How many terms of the sequence, $20 + 19\frac{1}{3} + 18\frac{2}{3} + \ldots$ must be taken so that their sum is 300. Explain the reason of double answer.

Sol. Sequence is AP with

$$a = 20 \text{ and } d = 19\frac{1}{3} - 20 = \frac{-2}{3}$$

let number of terms = n

hence $S_n = 300 = \frac{n}{2}[2a + (n-1)d]$

$$\Rightarrow \frac{n}{2}\left[40 - (n-1)\frac{2}{3}\right] = 300$$

$$\Rightarrow n\left(20 - \frac{(n-1)}{3}\right) = 300$$

$$\Rightarrow n(61-n) = 900 \Rightarrow n^2 - 61n + 900 = 0$$

$\Rightarrow (n - 36)(n - 25)=0 \Rightarrow n = 25, 36$

This is decreasing AP hence T_1 to T_{25} terms are positive. Hence $S_{25} = 300$. After this terms are getting reduced so sum will be increasing but when negative terms add up in the sequence, then sum will again get reduced and we get $S_{36} = 300$ again.

Q.7 The sum of n terms of two AP's are in the ratio of 7n+1 : 4n + 27, find the ratio of their 11th terms.

Sol. Given $\frac{S_n}{S'_n} = \frac{\frac{n}{2}[2a + (n-1)d]}{\frac{n}{2}[2a' + (n-1)d']} = \frac{7n+1}{4n+27}$

$$\Rightarrow \frac{2a + (n-1)d}{2a' + (n-1)d'} = \frac{7n+1}{4n+27}$$

$$\Rightarrow \frac{a+\frac{(n-1)}{2}d}{a'+\frac{(n-1)}{2}d'} = \frac{7n+1}{4n+27} \qquad \ldots(1)$$

for T_{11} we have $\frac{(n-1)}{2} = 10 \Rightarrow n = 21$

Put in (1) put n = 21

$$\frac{a+10d}{a'+10d'} = \frac{T_{11}}{T'_{11}} = \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3}$$

Q.8 If $S_1, S_2, S_3, \ldots, S_p$ are the sum of n terms of 'p' arithmetic series whose first terms are 1,2,3,4,... and whose common differences are 1,3,5,7,... prove that

$$S_1 + S_2 + S_3 + \ldots + S_p = \frac{np}{2}(np+1)$$

Sol. Clearly

$$S_1 = \frac{n}{2}\{2\times 1 + (n-1).1\}$$

$$S_2 = \frac{n}{2}\{2\times 2 + (n-1).3\}$$

$$S_3 = \frac{n}{2}\{2\times 3 + (n-1).5\}$$

$$\vdots$$

$$+S_p = \frac{n}{2}\{2\times p + (n-1).(2p-1)\}$$

$$S_1 + S_2 + \ldots S_p = \frac{n}{2}\{2.(1+2+3+\ldots+p) + (n-1)(1+3+5+\ldots+(2p-1))\}$$

$$= \frac{n}{2}\left\{2.\frac{p(p+1)}{2} + (n-1)p^2\right\}$$

$$= \frac{np}{2}\{np+1\} \text{ H.P.}$$

Q.9 In an AP $S_p = q$ and $S_q = p$ then show that $S_{p+q} = -(p+q)$

Sol. $S_p = q$

$$\Rightarrow \frac{p}{2}\{2a + (p-1)d\} = q$$

$$\Rightarrow ap + p(p-1)\frac{d}{2} = q \qquad \ldots(1)$$

also $S_q = p$

$$\Rightarrow \frac{q}{2}\{2a + (q-1)d\} = p$$

$$\Rightarrow aq + q(q-1)\frac{d}{2} = p \qquad \ldots(2)$$

(1) – (2)

$$\Rightarrow a(p-q) + \{(p^2 - q^2)-(p-q)\}\frac{d}{2} = -(p-q)$$

$$\Rightarrow a+(p+q-1)\frac{d}{2} = -1$$

$$\Rightarrow 2a+(p+q-1)d = -2 \qquad \ldots(3)$$

now

$$S_{p+q} = \frac{(p+q)}{2}\{2a + (p+q-1)d\}$$

$$= \frac{(p+q)}{2}(-2) = -(p+q) \qquad \text{H.P.}$$

Q.10 Let $a_1, a_2, a_3, \ldots, a_{100}$ be an arithmetric progression with $a_1 = 1$ and $S_p = \sum_{i=1}^{p} a_i$, $1 \le p \le 100$. For any integer n with $1 \le n \le 20$, let m = 5n. If $\frac{S_m}{S_n}$ does not depend on n then a_2 is.

Sol. $$\frac{S_{5n}}{S_n} = \frac{\frac{5n}{2}\{2a_1 + (5n-1)d\}}{\frac{n}{2}\{2a_1 + (n-1)d\}} = 5\left\{\frac{(2a_1 - d) + 5nd}{(2a_1 - d) + nd}\right\}$$

It is independent of n if

(i) d = 0 or (ii) $2a_1 = d \Rightarrow d = 2$

if d = 0 then $a_2 = a_1 = 1$

if d = 2 then $a_2 = a_1 + d = 3$

Q.11 The number of terms in an AP is even, the sum of the odd terms is 24, sum of the even terms is 30, and the last term exceeds the first term by $10\frac{1}{2}$.

Find the number of terms.

Sol. Let AP is $a_1, a_2, a_3, ..., a_{2n}$

sum of odd terms

$a_1 + a_3 + a_5 + ... + a_{2n-1} = 24$ ...(1)

sum of even terms

$a_2 + a_4 + a_6 + ... + a_{2n} = 30$ ...(2)

(2) - (1)

$(a_2 - a_1) + (a_4 - a_3) + ... + (a_{2n} - a_{2n-1}) = 6$

$nd = 6$ ... (3)

now given

$a_{2n} - a_1 = 10\frac{1}{2} = \frac{21}{2}$

$(2n-1)d = \frac{21}{2}$

$2nd - d = \frac{21}{2} \Rightarrow 2(6) - d = \frac{21}{2}$

hence $, d = \frac{3}{2}$

$n = \frac{6}{d} = \frac{6}{\frac{3}{2}} = 4$ (using 3)

So, number of terms = 2n = 8

Q.12 **Find the sum of all integers between 1 to 100 which are divisible by 2 or 3.**

Sol. Let the set

A = {2, 4, 6,...,98} divisible by 2

B = {3, 6, 9,...,99} divisible by 3

for set A

sum S_1 = 2+4+6+...+98

=2(1+2+3+...+49)

$= 2\left(\frac{49 \times 50}{2}\right) = 2450$

for set B

sum S_2 = 3+6+9+...+99

=3(1+2+3+...+33)

$= 3\left(\frac{33 \times 34}{2}\right) = 1683$

$A \cap B$ = {6, 12, 18, ..., 96}

Sum $S_3 = 6 + 12 + 18 + ... + 96$

$= 6(1 + 2 + 3 + ... + 16)$

$= 6\left(\frac{16 \times 17}{2}\right) = 816$

sum of all integers divisible by 2 or 3 $(A \cup B)$

$= S_1 + S_2 - S_3$

$= 2450 + 1683 - 816 = 3317$

Q.13 Find the nature and n^{th} term of the sequence whose sum upto n terms is $5n^2 + 2n + 4$

Sol. $S_n = 5n^2 + 2n + 4$

$T_n = S_n - S_{n-1}; n \geq 2$

$= \left(5n^2 + 2n + 4\right) - \left\{5(n-1)^2 + 2(n-1) + 4\right\}$

$= 5\left(n^2 - (n-1)^2\right) + 2(n - (n-1))$

$= 5(2n-1) + 2 = 10n - 3; n \geq 2$

hence sequence is

11, 17, 27, 37, ..., (where $T_1 = S_1$)

Q.14 Evaluate : $S = 100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + ... + 2^2 - 1^2$

Sol. $S = (100-99)(100+99) + (98-97)(98+97) + ... + (2-1)(2+1)$

$S = 100+99+98+97+...+2+1$

$= 100 \times \frac{101}{2} = 5050$

Q.15 Suppose $a_1, a_2, ...$ are in A.P and S_k denotes the sum of the first k terms of this AP. If $\frac{S_n}{S_m} = \frac{n^2}{m^2}$ for all m, $n \in N$, then prove that $\frac{a_{m+1}}{a_{n+1}} = \frac{(2m+1)}{(2n+1)}$

Sol. $\frac{S_n}{S_m} = \frac{(n/2)\left[2a_1 + (n-1)d\right]}{(m/2)\left[2a_1 + (m-1)d\right]} = \frac{n^2}{m^2}$

$\Rightarrow \frac{2a_1 + (n-1)d}{2a_1 + (m-1)d} = \frac{n}{m}$

$$\Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d}{a_1 + \left(\frac{m-1}{2}\right)d} = \frac{n}{m} \qquad \ldots(1)$$

In (1) replace $n \to 2n+1$, $m \to 2m+1$

$$\Rightarrow \frac{a_1 + nd}{a_1 + md} = \frac{(2n+1)}{(2m+1)}$$

Hence, $\frac{a_{m+1}}{a_{n+1}} = \frac{(2m+1)}{(2n+1)}$

Q.16 Let $a_1, a_2, \ldots, a_{30}$ be an AP, $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{(2i-1)}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to

(A) 57 (B) 47 (C) 42 (D) 52

Sol. **(D)**

$$S = \frac{30}{2}[2a_1 + (30-1)d]$$

$$S = [2a_1 + 29d]15 \qquad \ldots(1)$$

$$T = a_1 + a_3 + a_5 + \ldots + a_{29}$$

$$= \frac{15}{2}[2a_1 + (15-1)2d]$$

$$2T = 15[2a_1 + 28d] \qquad \ldots(2)$$

(1)–(2)

$S - 2T = 15(d) = 75 \Rightarrow d = 5$

Also $a_5 = 27$

$\Rightarrow a_1 + 4d = 27 \Rightarrow a_1 + 4(5) = 27$

$a_1 = 7$

$a_{10} = a_1 + 9d = 7 + 9(5) = 52$

Q.17 If a, b, c are in AP then prove that b + c, c + a, a + b are also in AP.

Sol. $\because$ a, b, c in AP

$\Rightarrow -a, -b, -c$ in AP

add (a + b + c) in all terms

$\Rightarrow$ (a + b + c) -a, (a + b + c)-b, (a + b + c)- c in AP

$\Rightarrow$ b + c, c + a, a + b in AP

Q.18 If a, b, c are in AP then prove that $(b+c)^2 - a^2, (c+a)^2 - b^2, (a+b)^2 - c^2$ are also in AP

Sol. $\because$ a, b, c in AP

$\Rightarrow -2a, -2b, -2c$ in AP

Add (a + b + c)

(b + c)−a, (c + a)−b, (a + b)−c in AP

Multiply by a + b + c

$(b+c)^2 - a^2, (c+a)^2 - b^2, (a+b)^2 - c^2$ in AP

Q.19 If a^2, b^2, c^2 are in AP then prove that $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in AP.

Sol. $\because$ a^2, b^2, c^2 in AP

add (ab + bc + ca) to each term

$\Rightarrow$ (a + b)(c + a), (a + b)(b + c), (b + c)(c + a) in AP

Divide by (a + b) (b + c) (c + a)

$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ in AP

Q.20 If a^2, b^2, c^2 are in AP then prove that $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in AP

Sol. $\because$ a^2, b^2, c^2 in AP

add (ab + bc + ca) to each term

$\Rightarrow$(a + b)(c + a), (a + b)(b + c), (b + c)(c + a) in AP

divide by (a + b)(b + c)(c + a)

$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ in AP

multiply by (a + b + c)

$\Rightarrow \frac{a+(b+c)}{b+c}, \frac{b+(c+a)}{c+a}, \frac{c+(a+b)}{a+b}$ in AP

$\Rightarrow \frac{a}{b+c}+1, \frac{b}{c+a}+1, \frac{c}{a+b}+1$ in AP

subtract 1

$\Rightarrow \frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ in AP

ARITHMETIC MEAN :

Consider n positive quantities $a_1, a_2, a_3, \ldots, a_n$.

$$\text{A.M.} = \frac{a_1 + a_2 + \ldots + a_n}{n}$$

Note: When three quantities are in AP. then the middle one is called the **Arithmetic Mean** of the other two.

To insert 'n' AM's between a and b.

Let $A_1, A_2, A_3, \ldots, A_n$ are n Arithmetic means between a and b

$b = a + (n + 2 - 1)d$

$$\Rightarrow d = \frac{b-a}{n+1}$$

$$\sum_{i=1}^{n} A_i = n\left[\frac{A_1 + A_n}{2}\right] = \frac{n}{2}[a+b]$$

= n times Arithmetic mean of 1st and last term.

Q.1 If 101 arithmetic means and are inserted between 1 and 99 then find their sum.

Sol. Sum of all arithmetic means

= 101 × (mean of 1 and 99)

$$= 101 \times \left(\frac{1+99}{2}\right)$$

= 101 × 50 = 5050

Q.2 Insert 20 AM's between 4 and 67.

Sol. 4, $A_1, A_2, A_3, \ldots, A_{20}$, 67 in AP

common difference $, d = \frac{67-4}{20+1} = \frac{63}{21} = 3$

$A_1 = 4 + d = 7$

$A_2 = 4 + 2d = 10$

⋮

$A_{20} = 4 + 20d = 64$

Q.3 If p arithmetic means are inserted between 5 and 41 so that the ratio $\frac{A_3}{A_{p-1}} = \frac{2}{5}$ then find the value of p.

Sol. 5, A_1, A_2, A_3, ..., Ap, 41

common difference $d = \frac{41-5}{p+1} = \frac{36}{p+1}$

$$\frac{A_3}{A_{p-1}} = \frac{2}{5} \Rightarrow 5A_3 = 2A_{p-1}$$

$$\Rightarrow 5(5+3d) = 2(5+(p-1)d)$$

$$\Rightarrow 25+15d = 10+(2p-2)d$$

$$\Rightarrow (2p-17)d = 15$$

$$\Rightarrow (2p-17)\frac{36}{p+1} = 15$$

$$\Rightarrow 12(2p-17) = 5(p+1)$$

$$\Rightarrow 19p = 209 \Rightarrow p = 11$$

Q.4 A number sequence $a_1, a_2, a_3, ..., a_n$ is such that

$a_1 = 0$; $|a_2| = |a_1 + 1|$; $|a_3| = |a_2 + 1|$; ... $|a_n| = |a_{n-1} + 1|$

Prove that the arithmetic mean of $a_1, a_2, ..., a_n$ is not less than $-\frac{1}{2}$

Sol. $|a_2| = |a_1 + 1|$

on squaring $a_2^2 = a_1^2 + 2a_1 + 1$

similarly, $a_3^2 = a_2^2 + 2a_2 + 1$

$$a_4^2 = a_3^2 + 2a_3 + 1$$

$$\vdots$$

$$a_{n+1}^2 = a_n^2 + 2a_n + 1$$

on adding all equations, we get

$$a_{n+1}^2 = a_1^2 + 2(a_1 + a_2 + ... + a_n) + n$$

$$\Rightarrow 2(a_1 + a_2 + ... + a_n) + n \geq 0$$

$$\Rightarrow \frac{a_1 + a_2 + ... + a_n}{n} \geq \frac{-1}{2}$$

Hence Proved.

General Illustration on A.P.

Q.5 If $\log_3 2, \log_3(2^x - 5)$ & $\log_3\left(2^x - \frac{7}{2}\right)$ are in AP determine x.

Sol. $2\log_3(2^x - 5) = \log_3 2 + \log_3\left(2^x - \frac{7}{2}\right)$

$\Rightarrow \log_3(2^x - 5)^2 = \log_3 2\left(2^x - \frac{7}{2}\right)$

$\Rightarrow (2^x - 5)^2 = 2\left(2^x - \frac{7}{2}\right)$

let $2^x = t$

$\Rightarrow (t-5)^2 = 2t - 7$

$\Rightarrow t^2 - 12t + 32 = 0$

$\Rightarrow (t-4)(t-8) = 0$

$\Rightarrow 2^x = 4\ ,\ 2^x = 8$

$\Rightarrow x = 2$ (rejected as $2^x - 5 < 0$), $x = 3$

$\Rightarrow x = 3$

Q.6 Solve the equation $\frac{x-1}{x} + \frac{x-2}{x} + \ldots + \frac{1}{x} = 3$

Sol. $\frac{1}{x}\{1 + 2 + 3 + \ldots + (x-1)\} = 3$

$\Rightarrow \frac{1}{x}\left\{\frac{(x-1)x}{2}\right\} = 3$

$\Rightarrow (x - 1) = 6 \Rightarrow x = 7$

Q.7 The fourth power of common difference of an arithmetic progression with integer entries is added to the product of any four consecutive terms of it prove that the resulting sum is the square of an integer.

Sol. Let 4 consecutive terms in AP are a - 3d, a - d, a + d, a + 3d where common difference = $2d \in I$

now required exp.

$= (a-3d)(a-d)(a+d)(a+3d) + (2d)^4$

$= (a^2 - 9d^2)(a^2 - d^2) + 16d^4$

$$= a^4 - 10a^2d^2 + 25d^4$$

$$= (a^2 - 5d^2)^2 = \{(a-3d)(a+3d) + (2d)^2\}^2$$

$$= (l_1l_2 + l_3^2) = (\text{Integer})^2$$

Q.8 Find the condition that the roots of equation $x^3 - px^2 + qx - r = 0$ may be in AP.

Sol. Let roots $\alpha - \beta, \alpha, \alpha + \beta$

Sum of roots = $3\alpha = p \Rightarrow \alpha = p/3$

$\because \alpha^3 - p\alpha^2 + q\alpha - r = 0$

$$\Rightarrow \frac{p^3}{27} - p.\frac{p^2}{9} + q.\frac{p}{3} - r = 0$$

$$\Rightarrow \frac{-2}{27}p^3 + \frac{1}{3}pq - r = 0$$

$$\Rightarrow 2p^3 - 9pq + 27r = 0 \text{ required condition}$$

Q.9 If the first 3 terms of an increasing AP are the roots of the cubic $4x^3 - 24x^2 + 23x + 18 = 0$ then find T_n.

Sol. Let roots are $\alpha - \beta, \alpha, \alpha + \beta$ (in AP) ($\beta > 0$)

Sum of roots = $3\alpha = \frac{24}{4} = 6 \Rightarrow \alpha = 2$

Product of roots = $\alpha(\alpha - \beta)(\alpha + \beta) = \frac{-18}{4}$

$$2(4 - \beta^2) = \frac{-9}{2}$$

$$\Rightarrow 4 - \beta^2 = \frac{-9}{4} \Rightarrow \beta^2 = \frac{25}{4}$$

$$\beta = \pm\frac{5}{2}$$

roots are $2 - \frac{5}{2}, 2, 2 + \frac{5}{2}$

$$\frac{-1}{2}, 2, \frac{9}{2}$$

for AP $a = \frac{-1}{2}, d = \frac{5}{2}$

$$T_n = \frac{-1}{2} + (n-1)\frac{5}{2} = \frac{5n-6}{2}$$

Q.10 If the sum of the roots of the equation $ax^2+bx+c=0$ is equal to the sum of the square of their reciprocals, then show that bc^2, ca^2, ab^2 are in AP.

Sol. Let roots α, β

now given that $\alpha+\beta=\frac{1}{\alpha^2}+\frac{1}{\beta^2}$

$$\Rightarrow \alpha+\beta=\frac{\alpha^2+\beta^2}{(\alpha\beta)^2}$$

$$\Rightarrow (\alpha+\beta)(\alpha\beta)^2=(\alpha+\beta)^2-2\alpha\beta$$

$$\Rightarrow -\frac{b}{a}\cdot\left(\frac{c}{a}\right)^2=\left(-\frac{b}{a}\right)^2-\frac{2c}{a}$$

$$\Rightarrow -\frac{bc^2}{a^3}=\frac{b^2}{a^2}-\frac{2c}{a} \Rightarrow -bc^2=ab^2-2a^2c$$

$$\Rightarrow ab^2+bc^2=2ca^2$$

hence bc^2, ca^2, ab^2 are in AP

Q.11 Given $a_1, a_2, a_3, \ldots, a_n$ are in AP. Prove that

$$\frac{1}{a_1a_n}+\frac{1}{a_2a_{n-1}}+\frac{1}{a_3a_{n-2}}+\ldots+\frac{1}{a_na_1}=\frac{2}{a_1+a_n}\left\{\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}\right\}$$

Sol. We know that $a_1+a_n=a_2+a_{n-1}=a_3+a_{n-2}=\ldots=k$ (let)

now LHS is

$$\Rightarrow \frac{1}{k}\left\{\frac{k}{a_1a_n}+\frac{k}{a_2a_{n-1}}+\frac{k}{a_3a_{n-2}}+\ldots+\frac{k}{a_na_1}\right\}$$

$$\Rightarrow \frac{1}{k}\left\{\left(\frac{1}{a_n}+\frac{1}{a_1}\right)+\left(\frac{1}{a_{n-1}}+\frac{1}{a_2}\right)+\left(\frac{1}{a_{n-2}}+\frac{1}{a_3}\right)+\ldots+\left(\frac{1}{a_1}+\frac{1}{a_n}\right)\right\}$$

$$\Rightarrow \frac{1}{k}2\left\{\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}\right\}=\frac{2}{a_1+a_n}\left\{\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}\right\}=\text{RHS}$$

Q.12 Let AP (a; d) denote the set of all the terms of an infinite arithmetic progression with first term a and common difference d > 0.
If $AP(1;3) \cap AP(2;5) \cap AP(3;7) = AP(a;d)$ then a+d equals.

Sol. Common terms of two or more than two APs also form an AP whose common difference is LCM of common differences of these APs

now

AP(1 ; 3) = {1,4,7,10,13,16,19,22,25,...}

AP(2 ; 5) = {2,7,12,17,22,27,32,37,42,...}

AP(3 ; 7) = {3,10,17,24,31,38,...}

first common terms in all these APs will be 52 hence a = 52

d = LCM {3, 5, 7} = 105

hence

a + d = 52 + 105 = 157

Q.13 Let $a_1, a_2, ..., a_n$ be a given AP whose common difference is an integer and $S_n = a_1 + a_2 + ... + a_n$. If $a_1 = 1, a_n = 300 \text{ and } 15 \le n \le 50$ then the ordered pair (S_{n-4}, a_{n-4}) is equal to

(A) (2480, 248) (B) (2480,249) (C) (2490,249) (D) (2490,248)

Sol. **(D)**

$a_n = a_1 + (n-1)d$

300 = 1+(n-1) d

Hence, $n - 1 = \frac{299}{d} \in [14, 49]$

$n - 1 = \frac{13 \times 23}{d} \in [14, 49]$

clearly d = 13 hence n = 24

$a_{n-4} = a_{20} = a_1 + 19d = 1 + 19(13) = 248$

and now

$S_{n-4} = S_{20} = \frac{20}{2}[a_1 + a_{20}] = 10(1 + 248) = 2490$

Q.14 Prove that $\sqrt{2}, \sqrt{3}, \sqrt{5}$ cannot be the terms of an AP (not necessarily adjacent)

Sol. Let these are p^{th}, q^{th}, r^{th} terms of an AP

$a + (p-1)d = \sqrt{2}$ ...(1)

$a+(q-1)d=\sqrt{3}$ ...(2)

$a+(r-1)d=\sqrt{5}$ ...(3)

(2)-(1), $(q-p)d=\sqrt{3}-\sqrt{2}$ ...(4)

(3)-(2), $(r-q)d=\sqrt{5}-\sqrt{3}$ ...(5)

now (4) ÷ (5)

$$\frac{q-p}{r-q}=\frac{\sqrt{3}-\sqrt{2}}{\sqrt{5}-\sqrt{3}}$$

LHS is rational as p, q, r ∈ N but RHS is irrational hence $\sqrt{2},\sqrt{3},\sqrt{5}$ can not be the terms of an AP.

GEOMETRIC PROGRESSION :

Definition

In a sequence if each term (except the first non- zero term) bears the same constant ratio with its immediately preceding term the sequence is called a **G.P.** and the constant ratio is called the **common ratio**.

Standard appearance of a GP is a, ar, ar^2, ar^3,..., ar^{n-1}.

n^{th} term = $T_n = ar^{n-1}\ (r \neq 0)$

Q.1 In a G.P. if $t_3 = 2$ and $t_6 = -\frac{1}{4}$, find t_{10}

Sol. ∵ $t_3 = 2 \Rightarrow ar^2 = 2$ ...(1)

also $t_6 = -\frac{1}{4} \Rightarrow ar^5 = -\frac{1}{4}$ ...(2)

(2) ÷ (1)

$$r^3 = -\frac{1}{8} \Rightarrow r = -\frac{1}{2}$$

$$a = \frac{2}{r^2} = \frac{2}{\left(\frac{1}{4}\right)} = 8$$

$$t_{10} = ar^9 = 8\left(-\frac{1}{2}\right)^9 = -\frac{1}{64}$$

Q.2 If p^{th}, q^{th} and r^{th} terms of a G.P. are x, y and z respectively then prove that $x^{q-r} \cdot y^{r-p} \cdot z^{p-q} = 1$

Sol. Given

$$T_p = x \Rightarrow AR^{p-1} = x \quad \ldots(1)$$

$$T_q = y \Rightarrow AR^{q-1} = y \quad \ldots(2)$$

$$T_r = z \Rightarrow AR^{r-1} = z \quad \ldots(3)$$

$$\text{now } x^{q-r} = \left(AR^{p-1}\right)^{q-r} = A^{q-r}R^{(p-1)(q-r)}$$

$$= A^{q-r}R^{pq-pr-q+r} \quad \ldots(4)$$

$$y^{r-p} = \left(AR^{q-1}\right)^{r-p} = A^{r-p}R^{(q-1)(r-p)}$$

$$= A^{r-p}R^{qr-pq-r+p} \quad \ldots(5)$$

$$z^{p-q} = \left(AR^{r-1}\right)^{p-q} = A^{p-q}R^{(r-1)(p-q)}$$

$$= A^{p-q}R^{pr-qr-p+q} \quad \ldots(6)$$

(4) × (5) × (6)

$$x^{q-r}.y^{r-p}.z^{p-q} = A^{\circ}R^{\circ} = 1$$

Sum of n terms of a G. P.

$$\mathbf{S = a + ar + ar^2 + \ldots + ar^{n-1}}$$

$$S = \frac{a\left(1-r^n\right)}{1-r}$$

where $r \neq 1$, If $r = 1$ then $S = na$

If $|r| < 1$ and $n \to \infty$ then

$$S_{\infty} = \frac{a}{1-r}$$

Points to Remember!!!

(i) If each term of a GP be multiplied or divided by the same non-zero quantity, the resulting sequence is also a GP.

(ii) Any 3 consecutive terms of a GP can be taken as $\frac{a}{r}, a, ar$.

Any 4 consecutive terms of a GP can be taken as $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$ and so on.

(iii) If a, b, c are in GP $\Rightarrow$ **$b^2 = ac$**

Q.1 The sum of first 3 consecutive terms of a GP is 19 and their product is 216. find S_n. Also compute S_∞ if it exists.

Sol. Let terms are $\alpha/\beta, \alpha, \alpha\beta$

Given product = $\frac{\alpha}{\beta}.\alpha.\alpha\beta = 216$

$\Rightarrow \alpha^3 = 6^3 \Rightarrow \alpha = 6$

also sum $\Rightarrow \frac{\alpha}{\beta} + \alpha + \alpha\beta = 19$

put $\alpha = 6$ then $\frac{6}{\beta} + 6 + 6\beta = 19$

$\Rightarrow 6 + 6\beta + 6\beta^2 = 19\beta$
$\Rightarrow 6\beta^2 - 13\beta + 6 = 0$
$\Rightarrow 6\beta^2 - 9\beta - 4\beta + 6 = 0$
$\Rightarrow 3\beta(2\beta-3) - 2(2\beta-3) = 0$
$\Rightarrow (3\beta-2)(2\beta-3) = 0$

$\Rightarrow \beta = \frac{2}{3}$ or $\frac{3}{2}$

hence numbers

for $\beta = \frac{2}{3} \Rightarrow \frac{6}{\frac{2}{3}}, 6, 6.\left(\frac{2}{3}\right) \Rightarrow 9, 6, 4$

for $\beta = \frac{3}{2} \Rightarrow \frac{6}{\frac{3}{2}}, 6, 6.\left(\frac{3}{2}\right) \Rightarrow 4, 6, 9$

Case–I: GP 9, 6, 4, ...

A = 9, r = 2/3

$$S_n = 9\left\{\frac{1-\left(\frac{2}{3}\right)^n}{1-\frac{2}{3}}\right\} = 27\left\{1-\left(\frac{2}{3}\right)^n\right\}$$

$$S_\infty = \frac{9}{1-\frac{2}{3}} = 27$$

Case–II: GP 4, 6, 9, ...

$$a = 4, r = \frac{3}{2}$$

$$S_n = 4\left\{\frac{\left(\frac{3}{2}\right)^n - 1}{\frac{3}{2} - 1}\right\} = 8\left\{\left(\frac{3}{2}\right)^n - 1\right\}$$

S_∞ will not exist as $r = \frac{3}{2} > 1$

Q.2 In a GP, find a_n if $a_1 + a_2 + a_3 = 13$ and $a_1^2 + a_2^2 + a_3^2 = 91$

Sol. Let $a_2 = a_1 r, a_3 = a_1 r^2$

now $a_1 + a_2 + a_3 = 13$

$$\Rightarrow a_1 + a_1 r + a_1 r^2 = 13$$

$$\Rightarrow a_1\left(1 + r + r^2\right) = 13 \qquad \ldots(1)$$

Also, $a_1^2 + a_2^2 + a_3^2 = 91$

$$\Rightarrow a_1^2 + a_1^2 r^2 + a_1^2 r^4 = 91$$

$$\Rightarrow a_1^2\left(1 + r^2 + r^4\right) = 91 \qquad \ldots(2)$$

(2) ÷ $(1)^2$

$$\Rightarrow \frac{a_1^2\left(1 + r + r^2\right)\left(1 - r + r^2\right)}{a_1^2\left(1 + r + r^2\right)^2} = \frac{13 \times 7}{13^2}$$

$$\Rightarrow \frac{1 - r + r^2}{1 + r + r^2} = \frac{7}{13}$$

$$\Rightarrow 13 - 13r + 13r^2 = 7 + 7r + 7r^2$$

$$\Rightarrow 6r^2 - 20r + 6 = 0$$

$$\Rightarrow 3r^2 - 10r + 3 = 0$$

$$\Rightarrow (3r - 1)(r - 3) = 0 \Rightarrow r = \frac{1}{3}, r = 3$$

from (1)

if $r = \frac{1}{3} \Rightarrow a_1\left(1 + \frac{1}{3} + \frac{1}{9}\right) = 13 \Rightarrow a_1 = 9$

if $r = 3 \Rightarrow a_1(1 + 3 + 9) = 13 \Rightarrow a_1 = 1$

Case–I: GP $9, \frac{9}{3}, \frac{9}{3^2}, \ldots$

$$a_n = 9\left(\frac{1}{3}\right)^{n-1}$$

Case–II: GP $1, 3, 3^2, \ldots$

$$a_n = 1(3)^{n-1}$$

Q.3 If $\sin x$, $\sin^2 2x$ and $\cos x \cdot \sin 4x$ form an increasing geometric sequence, find the numerical value of $\cos^2 2x + \cos 2x$. Also find the common ratio of geometric sequence.

Sol. As $\sin x, \sin^2 2x, \cos x \sin 4x$ are in GP

$$\Rightarrow \left(\sin^2 2x\right)^2 = (\sin x).(\cos x \sin 4x)$$

$$\Rightarrow \left(\sin^2 2x\right)^2 = (\sin x \cos x)(2\sin 2x \cos 2x)$$

$$\Rightarrow \left(\sin^2 2x\right)^2 = \left(\sin^2 2x\right)(\cos 2x)$$

$$\Rightarrow \left(\sin^2 2x\right)\left\{\sin^2 2x - \cos 2x\right\} = 0$$

$\Rightarrow \sin 2x = 0$ (not possible), $1 - \cos^2 2x - \cos 2x = 0$

$\Rightarrow \cos^2 2x + \cos 2x = 1$ ... (1)

$$(\text{common ratio})^2 = \frac{\cos x \ \sin 4x}{\sin x}$$

$$r^2 = \frac{\cos x \ 2\sin 2x \ \cos 2x}{\sin x}$$

$$= \frac{\cos x \ 2(2\sin x \cos x)\cos 2x}{\sin x}$$

$$= 4\cos^2 x \cos 2x = 2(1 + \cos 2x)\cos 2x$$

$$= 2\left(\cos^2 2x + \cos 2x\right) = 2(1) \quad \text{[Using (1)]}$$

$\Rightarrow r = \sqrt{2}$ (as GP is increasing)

Q.4 The sum of an infinite number of terms of a GP is 15 and the sum of their squares is 45. Find the series.

Sol. Let GP be $a, ar, ar^2, \ldots$

sum of infinite terms $= \frac{a}{1-r} = 15$

$\Rightarrow a = 15(1-r)$ ...(1)

squared sequence: $a^2, a^2r^2, a^2r^4, \ldots$

sum $= \frac{a^2}{1-r^2} = 45 \Rightarrow a^2 = 45(1+r)(1-r)$ ...(2)

$(1)^2 \div (2)$

$$1 = \frac{225(1-r)^2}{45(1+r)(1-r)}$$

$(1+r) = 5(1-r) \Rightarrow 6r = 4$

$r = \frac{2}{3}$

from (1) $\Rightarrow a = 15\left(1-\frac{2}{3}\right) = 5$

Hence, series is

$5, 5.\frac{2}{3}, 5.\left(\frac{2}{3}\right)^2, \ldots$

$\Rightarrow 5, \frac{10}{3}, \frac{20}{9}, \ldots$

Q.5 If α and β are the roots of equation $375x^2 - 25x - 2 = 0$, then $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\alpha^r + \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\beta^r$ is equal to

(A) $\frac{21}{346}$ (B) $\frac{29}{358}$ (C) $\frac{1}{12}$ (D) $\frac{7}{116}$

Sol. (C)

$\alpha + \beta = \frac{25}{375} = \frac{1}{15}$, $\alpha\beta = \frac{-2}{375}$

Now given exp. is $\sum\limits_{r=1}^{\infty}\alpha^r + \sum\limits_{r=1}^{\infty}\beta^r$

$\Rightarrow (\alpha + \alpha^2 + \alpha^3 + \ldots) + (\beta + \beta^2 + \beta^3 + \ldots)$

$$\Rightarrow \frac{\alpha}{1-\alpha}+\frac{\beta}{1-\beta} \Rightarrow \frac{\alpha(1-\beta)+\beta(1-\alpha)}{(1-\alpha)(1-\beta)}$$

$$\Rightarrow \frac{(\alpha+\beta)-2\alpha\beta}{1-(\alpha+\beta)+\alpha\beta} \Rightarrow \left\{\frac{25}{375}+\frac{2(2)}{375}\right\} \times \frac{1}{\left(1-\frac{25}{375}-\frac{2}{375}\right)}$$

$$\Rightarrow \frac{29}{375} \times \frac{375}{348} = \frac{29}{348} = \frac{1}{12}$$

Q.6 Use infinite series to compute the rational number corresponding to $0.4\overline{23}$

Sol. Let $x = 0.4232323...$

$x = 0.4 + 0.023 + 0.00023 + ...\infty$

$$x = \frac{4}{10}+\frac{23}{10^3}+\frac{23}{10^5}+\frac{23}{10^7}+...\infty$$

$$x = \frac{2}{5}+\frac{\frac{23}{10^3}}{1-\frac{1}{10^2}}$$

$$x = \frac{2}{5}+\frac{23}{10^3}\cdot\frac{10^2}{99} = \frac{2}{5}+\frac{23}{990} = \frac{419}{990}$$

Q.7 Find the sum : $S = 9 + 99 + 999 + ...+ 999...9$(n times)

Sol. $S = (10-1)+(10^2-1)+(10^3-1)+...+(10^n-1)$

$S = (10+10^2+10^3+...+10^n)-n$

$$S = \frac{10(10^n-1)}{9} - n$$

$$S = \frac{1}{9}(10^{n+1}-9n-10)$$

Q.8 Find the sum : $S = 0.9 + 0.99 + 0.999 + ...$ upto n times

Sol. $$S = \frac{9}{10}+\frac{99}{10^2}+\frac{999}{10^3}+...$$

$$S = \frac{(10-1)}{10}+\frac{(10^2-1)}{10^2}+\frac{(10^3-1)}{10^3}+...+\frac{(10^n-1)}{10^n}$$

$$S = \left(1-\frac{1}{10}\right)+\left(1-\frac{1}{10^2}\right)+\left(1-\frac{1}{10^3}\right)+...+\left(1-\frac{1}{10^n}\right)$$

$$S = n - \left(\frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^3} + \ldots + \frac{1}{10^n}\right)$$

$$S = n - \frac{\frac{1}{10}\left(1 - \frac{1}{10^n}\right)}{\left(1 - \frac{1}{10}\right)} = n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)$$

Q.9 Solve the following equations for x and y

$\log_2 x + \log_4 x + \log_{16} x + \ldots = y$

$$\frac{5 + 9 + 13 + \ldots + (4y + 1)}{1 + 3 + 5 + \ldots + (2y - 1)} = 4\log_4 x$$

Sol. Given equation

$$\log_2 x + \frac{1}{2}\log_2 x + \frac{1}{4}\log_2 x + \ldots = y$$

$$\Rightarrow \log_2 x.\left\{1 + \frac{1}{2} + \frac{1}{4} + \ldots\right\} = y$$

$$\Rightarrow \log_2 x.\left\{\frac{1}{1 - \frac{1}{2}}\right\} = y$$

$$\Rightarrow \log_2 x = \frac{y}{2} \qquad \ldots(1)$$

Second equation

$$\frac{\frac{y}{2}\{5 + (4y + 1)\}}{y^2} = 4.\frac{1}{2}\log_2 x$$

$$\Rightarrow y(3 + 2y) = 2\log_2 x \cdot y^2$$

$$\Rightarrow 2y^2 + 3y = 2.\frac{y}{2}.y^2 \Rightarrow y^3 - 2y^2 - 3y = 0$$

$$\Rightarrow y\left(y^2 - 2y - 3\right) = 0 \Rightarrow y(y + 1)(y - 3) = 0$$

$\Rightarrow y = 0, -1, 3$

As $y \in N$, So, $y = 3$

from (1) $\log_2 x = \frac{3}{2} \Rightarrow x = 2^{\frac{3}{2}} = 2\sqrt{2}$

Q.10 For any three positive real numbers a, b and c,
$9(25a^2+b^2)+25(c^2-3ac)=15b(3a+c)$ then

(A) b, c and a are in GP (B) b, c and a are in AP
(C) a, b and c are in AP (D) a, b and c are in GP

Sol. **(B)**

Given equation is

$$(15a)^2+(3b)^2+(5c)^2-(15a)(3b)-(3b)(5c)-(15a)(5c)=0$$

$$\Rightarrow 15a=3b=5c=k\,(\text{let})$$

$$a=\frac{k}{15},b=\frac{k}{3},c=\frac{k}{5}$$

$$a+b=\frac{k}{3}\left(\frac{1}{5}+1\right)=\frac{k}{3}.\frac{6}{5}=\frac{2k}{5}=2c$$

$\Rightarrow$ b, c, a in AP

Q.11 If a, b and c be three distinct real numbers in GP and a + b + c = xb then x cannot be

(A) 4 (B) −3 (C) −2 (D) 2

Sol. **(D)**

Let $b = ar$, $c = ar^2$ $(r \neq \pm 1)$

Now, $a + b + c = xb$

$\Rightarrow a + ar + ar^2 = x.ar$

$\Rightarrow 1 + r + r^2 = xr$ ...(1)

$\Rightarrow r^2+(1 - x)r + 1=0$

Now $D \geq 0$

$$\Rightarrow (1-x)^2-4(1)\geq 0$$

$$\Rightarrow (1-x+2)(1-x-2)\geq 0$$

$$\Rightarrow (x-3)(x+1)\geq 0$$

$$\Rightarrow x\in(-\infty,-1]\cup[3,\infty)$$

from (1) if $r = 1 \Rightarrow x = 3$

if $r = -1 \Rightarrow x = -1$

clearly, $x \in (-\infty, -1) \cup (3, \infty)$

Q.12 Find four successive terms of a GP of which the 2nd term is smaller than the first by 35 and the 3rd term is larger than the 4th by 560.

Sol. Let terms are a, ar, ar^2, ar^3
given $a - ar = 35$

$$a(1-r) = 35 \quad \ldots(1)$$

and $ar^2 - ar^3 = 560$

$$ar^2(1-r) = 560 \quad \ldots(2)$$

(2)÷(1)

$$r^2 = \frac{560}{35} = 16 \Rightarrow r = \pm 4$$

If r = 4 then $a = -\frac{35}{3}$

Terms are $\frac{-35}{3}, \frac{-140}{3}, \frac{-560}{3}, \frac{-2240}{3}$

If r = − 4 then $a = \frac{35}{5} = 7$

Terms are 7, −28, 112, −448

Q.13 If $p^{th}, q^{th}, r^{th}, s^{th}$ terms of an AP are in GP show that p-q, q-r, r-s are in GP.

Sol. Let

$$a + (p-1)d = A \quad \ldots(1)$$

$$a + (q-1)d = AR \quad \ldots(2)$$

$$a + (r-1)d = AR^2 \quad \ldots(3)$$

$$a + (s-1)d = AR^3 \quad \ldots(4)$$

(1) − (2) ⇒ (p − q)d = A(1-R)
(2) − (3) ⇒ (q − r)d = AR(1-R)
(3) − (4) ⇒ (r − s)d = AR^2(1-R)
Clearly, p − q, q − r, r − s are in GP

Q.14 If loga, logb, logc are in AP then prove that a, b, c are in GP.

Sol. Given loga, logb, logc in AP

$$\Rightarrow 2\log b = \log a + \log c$$
$$\Rightarrow \log b^2 = \log ac$$
$$\Rightarrow b^2 = ac$$
$$\Rightarrow a, b, c \text{ are in GP}$$

Q.15 If $S_1, S_2, S_3, \ldots, S_n, \ldots$ are the sums of infinite geometric series whose first terms are 1,2,3, ..., n, ... and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n+1}, \ldots$ respectively then find the value of $\sum_{r=1}^{2n-1} S_r^2$

Sol. For $S_1; a = 1, r = \frac{1}{2}$

$$\Rightarrow S_1 = \frac{1}{1-\frac{1}{2}} = 2$$

for $S_2; a = 2, r = \frac{1}{3} \Rightarrow S_2 = \frac{2}{1-\frac{1}{3}} = 3$

Similarly, $S_3 = 4, S_4 = 5, \ldots S_r = (r+1)$

$$\sum_{r=1}^{2n-1} S_r^2 = \sum_{r=1}^{2n-1} (r+1)^2 = \sum_{r=1}^{2n} r^2 - 1$$

$$= \frac{(2n)(2n+1)(4n+1)}{6} - 1$$

$$= \frac{n}{3}(2n+1)(4n+1) - 1$$

GEOMETRIC MEAN :

Consider n positive quantities $a_1, a_2, a_3, \ldots, a_n$.

G.M. = $(a_1 \cdot a_2 \cdot \ldots \cdot a_n)^{1/n}$

To insert 'n' GM's between a and b

Let a and b are two positive numbers and $G_1, G_2, \ldots, G_n$ are 'n' GM's then

$$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

here $\prod_{i=1}^{n} G_i = (G)^n$

where G is G.M. between a and b

Hence product of n GM's inserted between a and b is equal to the nth power of single GM between them. It may be noted that between two positive numbers

AM ≥ GM

Definition

If a, b, c are three positive numbers in GP then b is called **geometric mean** between a and c and **$b^2 = ac$.**

Q.1 Insert 4 GM's between 5 and 160.

Sol. Let $5, G_1, G_2, G_3, G_4, 160 \rightarrow$ in GP

$$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}} = \left(\frac{160}{5}\right)^{\frac{1}{5}} = (32)^{\frac{1}{5}} = 2$$

$G_1 = 5r = 10, G_3 = 5r^3 = 40,$ $\quad G_2 = 5r^2 = 20, G_4 = 5r^4 = 80$

Q.2 If AM between a and b is 15 and GM between a and b is 9. Find the number.

Sol. $\frac{a+b}{2} = 15 \Rightarrow a + b = 30$ ...(1)

$\sqrt{ab} = 9 \Rightarrow ab = 81$ ...(2)

by (1) and (2) $a + \frac{81}{a} = 30$

$$a^2 - 30a + 81 = 0 \Rightarrow (a - 27)(a - 3) = 0$$

$a = 3, 27$

If $a = 3$ then $b = 27$

$a = 27$ then $b = 3$

Q.3 If sum of two numbers a and b (a > b) is n(n > 2) times their GM then show that $a : b = n + \sqrt{n^2 - 4} : n - \sqrt{n^2 - 4}$

Sol. Given AM = n GM

$$a + b = n\sqrt{ab} \Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{n}{2}$$

by using componendo and dividendo

$$\frac{a + b + 2\sqrt{ab}}{a + b - 2\sqrt{ab}} = \frac{n+2}{n-2}$$

$$\Rightarrow \left(\frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}}\right)^2 = \frac{n+2}{n-2}$$

$$\Rightarrow \frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} = \frac{\sqrt{n+2}}{\sqrt{n-2}}$$

Again, by using C and D

$$\Rightarrow \frac{\sqrt{a}}{\sqrt{b}} = \frac{\sqrt{n+2} + \sqrt{n-2}}{\sqrt{n+2} - \sqrt{n-2}}$$

on Squaring both sides

$$\frac{a}{b} = \frac{2n + 2\sqrt{n^2 - 4}}{2n - 2\sqrt{n^2 - 4}} = \frac{n + \sqrt{n^2 - 4}}{n - \sqrt{n^2 - 4}}$$

Hence proved.

Q.4 If a, b, c are in GP and x, y are respectively the AM's between a, b and b, c respectively, then prove that

$$\frac{1}{x}+\frac{1}{y}=\frac{2}{b} \text{ and } \frac{a}{x}+\frac{c}{y}=2$$

Sol. Let $b = ar, c = ar^2$

and $x = \frac{a+b}{2} = \frac{a}{2}(1+r)$

$y = \frac{b+c}{2} = \frac{ar}{2}(1+r)$

now, $\frac{1}{x}+\frac{1}{y} = \frac{2}{a(1+r)}+\frac{2}{ar(1+r)} = \frac{2}{ar(1+r)}(r+1) = \frac{2}{ar} = \frac{2}{b}$

now, $\frac{a}{x}+\frac{c}{y} = \frac{2a}{a(1+r)}+\frac{2ar^2}{ar(1+r)}$

$= \frac{2}{1+r}+\frac{2r}{1+r} = 2$ **Hence proved.**

Q.5 If a is the AM of b and c and the two geometric means between b and c are G_1 and G_2 then prove that $G_1^3 + G_2^3 = 2abc$

Sol. Clearly, $a = \frac{b+c}{2} \Rightarrow b + c = 2a$ ...(1)

now G_1 and G_2 are GM's between b and c

$\Rightarrow$ $b, G_1, G_2, C \rightarrow$ in GP

$\Rightarrow$ $G_1^2 = bG_2$ and $G_2^2 = G_1c$ and $G_1G_2 = bc$

$G_1^3 = bG_1G_2$ and $G_2^3 = G_1G_2c$

$G_1^3 = b(bc)$ and $G_2^3 = (bc)c$

$G_1^3 + G_2^3 = b^2c + bc^2 = bc(b + c) = 2abc$ **Hence proved.**

Q.6 If a, b, c, d are four positive real numbers such that abcd = 1.
Prove that (1 + a) (1 + b) (1 + c) (1 + d) ≥ 16

Sol. ∵ AM ≥ GM

$\Rightarrow \frac{1+a}{2} \geq \sqrt{a} \Rightarrow 1 + a \geq 2\sqrt{a}$ ...(1)

Similarly

$1 + b \geq 2\sqrt{b}$

$1 + c \geq 2\sqrt{c}$

$1 + d \geq 2\sqrt{d}$

on multiplying we get

$(1 + a)(1 + b)(1 + c)(1 + d) \geq 16\sqrt{abcd}$

$\Rightarrow (1 + a)(1 + b)(1 + c)(1 + d) \geq 16$ **Hence Proved.**

Q.7 If $x > 0, y > 0, z > 0$ then prove that $(x + y)(y + z)(z + x) \geq 8xyz$

Sol. $AM \geq GM \Rightarrow \frac{x+y}{2} \geq \sqrt{xy}$

$\Rightarrow x + y \geq 2\sqrt{xy}$

Similarly, $y + z \geq 2\sqrt{yz}$

$z + x \geq 2\sqrt{zx}$

on multiplying we get

$\Rightarrow (x + y)(y + z)(z + x) \geq 8xyz$ **Hence proved.**

Q.8 If the sum of first 20 terms of the series $\log_{\frac{1}{7^3}} x + \log_{\frac{1}{7^3}} x + \log_{\frac{1}{7^4}} x + \ldots$ is 460, then x is equal to

(A) $7^{\frac{1}{2}}$ (B) 7^2 (C) e^2 (D) $7^{\frac{46}{21}}$

Sol. **(B)**

$\text{sum} = 2\log_7 x + 3\log_7 x + 4\log_7 x + \ldots$

$\Rightarrow \log_7 x\{2 + 3 + 4 + \ldots + 21\} = 460$

$\Rightarrow (\log_7 x)\left(\frac{21 \times 22}{2} - 1\right) = 460$

$\Rightarrow (\log_7 x)(230) = 460 \Rightarrow \log_7 x = 2 \Rightarrow x = 7^2$

Q.9 Prove that a Δ ABC is equilateral if and only if $\tan A + \tan B + \tan C = 3\sqrt{3}$

Sol. Let Δ ABC is equilateral triangle then $A = B = C = 60°$

$\Rightarrow \tan A + \tan B + \tan C = 3\tan 60° = 3\sqrt{3}$

Conversely,

Let $\tan A + \tan B + \tan C = 3\sqrt{3}$

$\Rightarrow \tan A \cdot \tan B \cdot \tan C = 3\sqrt{3}$

$\Rightarrow$ none of tanA, tanB, tanC can be negative so applying AM $\geq$ GM

$\frac{\tan A + \tan B + \tan C}{3} \geq \{\tan A.\tan B.\tan C\}^{\frac{1}{3}}$

$\because \tan A + \tan B + \tan C = \tan A \,.\, \tan B \,.\, \tan C = \alpha$(let)

$$\Rightarrow \frac{\alpha}{3} \geq \alpha^{\frac{1}{3}} \Rightarrow \frac{\alpha^3}{27} \geq \alpha \Rightarrow \alpha^2 \geq 27$$

$$\alpha \geq 3\sqrt{3}$$

So, the equality holds if and only if
tanA = tanB = tanC
$\Rightarrow$ A= B = C $\Rightarrow$ ΔABC is equilateral. **Hence Proved.**

Q.10 If a + b + c = 3 and a, b, c are positive then prove that $a^2b^3c^2 \leq \frac{3^{10}.2^4}{7^7}$

Sol. Using AM $\geq$ GM

$$\Rightarrow \frac{\frac{a}{2}+\frac{a}{2}+\frac{b}{3}+\frac{b}{3}+\frac{b}{3}+\frac{c}{2}+\frac{c}{2}}{7} \geq \left(\frac{a^2b^3c^2}{2^4 3^3}\right)^{\frac{1}{7}}$$

$$\Rightarrow \left(\frac{a+b+c}{7}\right)^7 \geq \frac{a^2b^3c^2}{2^4 3^3} \Rightarrow \frac{3^7}{7^7} \times 2^4 \times 3^3 \geq a^2b^3c^2$$

$$\Rightarrow a^2b^3c^2 \leq \frac{3^{10}2^4}{7^7}$$ **Hence Proved.**

Q.11 If n is positive integer, show that $2^{2n+1} > 1+(2n+1)2^n$

Sol. Using AM $\geq$ GM

$$\frac{1+2+2^2+2^3+\ldots+2^{2n}}{(2n+1)} > \left(1.2.2^2\ldots2^{2n}\right)^{\frac{1}{2n+1}}$$

(Here numbers are unequal hence equality will not hold)

$$\frac{1\left(2^{2n+1}-1\right)}{(2n+1)} > \left(2^{1+2+3+\ldots+2n}\right)^{\frac{1}{2n+1}}$$

$$\Rightarrow 2^{2n+1}-1 > (2n+1)\left(2^{\frac{2n(2n+1)}{2}}\right)^{\frac{1}{2n+1}}$$

$$\Rightarrow 2^{2n+1} > 1+(2n+1)\left(2^n\right)$$ **Hence proved.**

Q.12 In the equation $x^4 + px^3 + qx^2 + rx + 5 = 0$ has four positive real roots then find the minimum value of pr.

Sol. Let roots are a, b, c and d
then a + b + c + d = – p
ab + ac + ad + bc + bd + cd = q
abc + abd + acd + bcd = - r
abcd = 5
$\because$ AM $\geq$ GM

$$\therefore \frac{a+b+c+d}{4} \geq (abcd)^{\frac{1}{4}}$$

$\Rightarrow a + b + c + d \geq 4(5)^{1/4}$

$\Rightarrow -p \geq 4(5)^{1/4}$ ...(1)

also $\frac{abc+abd+acd+bcd}{4} \geq (a^3b^3c^3d^3)^{\frac{1}{4}}$

$\Rightarrow abc + abd + acd + bcd \geq 4(5)^{3/4}$

$\Rightarrow -r \geq 4(5)^{3/4}$ ...(2)

by (1) & (2)

$((-p)(-r))_{min} = 4(5^{1/4}).4(5^{3/4}) = 16(5)$

$pr_{min} = 80$

Q.13 If a, b, c are positive real numbers, then prove that $[(1+a)(1+b)(1+c)]^7 > 7^7a^4b^4c^4$

Sol. Using AM ≥ GM

$$\frac{a+b+c+ab+bc+ca+abc}{7} \geq (a^4b^4c^4)^{\frac{1}{7}}$$

$$\frac{1+a+b+c+ab+bc+ca+abc}{7} > (a^4b^4c^4)^{\frac{1}{7}}$$

$\Rightarrow [(1+a)(1+b)(1+c)]^7 > 7^7a^4b^4c^4$ **Hence Proved.**

ARITHMETIC GEOMETRIC PROGRESSION (AGP)

Standard appearance of an AGP is

$$S = a + (a+d)r + (a+2d)r^2 + (a+3d)r^3 + \ldots$$

Q.1 If $|x| < 1$ then compute the sum : $1 + 2x + 3x^2 + 4x^3 + \ldots\infty$

Sol. Let

$$S = 1 + 2x + 3x^2 + 4x^3 + \ldots$$
$$xS = \quad x + 2x^2 + 3x^3 + \ldots$$
$$- \quad - \quad - \quad - \quad -$$
$$(1-x)S = 1 + x + x^2 + x^3 + \ldots$$

$$(1-x)S = \frac{1}{1-x} \Rightarrow S = \frac{1}{(1-x)^2} = (1-x)^{-2}$$

Q.2 If $|x| < 1$ then compute the sum : $1 + 3x + 6x^2 + 10x^3 + \ldots\infty$

Sol. Let

$$S = 1 + 3x + 6x^2 + 10x^3 + \ldots$$

$$xS = \quad x + 3x^2 + 6x^3 + \ldots$$

$$(1-x)S = 1 + 2x + 3x^2 + 4x^3 + \ldots$$

$(1-x)S = (1-x)^{-2}$ (from previous question)

$S = (1-x)^{-3}$

Q.3 Find the sum to n terms and also S_∞ of series $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots$

Sol. Let

$$S_n = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots + \frac{3n-2}{5^{n-1}}$$

$$\frac{S_n}{5} = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + \quad \ldots \quad + \frac{3n-2}{5^n}$$

$$4\frac{S_n}{5} = 1 + \frac{3}{5} + \frac{3}{5^2} + \frac{3}{5^3} + \ldots - \frac{(3n-2)}{5^n}$$

$$4\frac{S_n}{5} = 1 + \frac{3}{5}\left(1 + \frac{1}{5} + \frac{1}{5^2} + \ldots (n-1)\text{ terms}\right) - \frac{(3n-2)}{5^n}$$

$$4\frac{S_n}{5} = 1 + \frac{3}{5}\left\{\frac{1-\left(\frac{1}{5}\right)^{n-1}}{1-\frac{1}{5}}\right\} - \frac{(3n-2)}{5^n}$$

$$4\frac{S_n}{5} = 1 + \frac{3}{4}\left\{1 - \frac{1}{5^{n-1}}\right\} - \left(\frac{3n-2}{5^n}\right)$$

$$S_n = \frac{5}{4} + \frac{15}{16}\left(1 - \frac{1}{5^{n-1}}\right) - \frac{1}{4}\left(\frac{3n-2}{5^{n-1}}\right)$$

when $n \to \infty$ then $\frac{1}{5^{n-1}} = 0$ and also $\frac{1}{4}\frac{3n-2}{5^{n-1}} = 0$

hence $S_\infty = \frac{5}{4} + \frac{15}{16} = \frac{35}{16}$

Q.4 Find S_∞ for $\frac{3}{5}+\frac{5}{15}+\frac{7}{45}+\frac{9}{135}+\ldots$

Sol. Let

$$S_n=\frac{3}{5}+\frac{5}{15}+\frac{7}{45}+\frac{9}{135}+\ldots+\frac{2n+1}{5(3)^{n-1}}$$

$$\frac{S_n}{3}=\quad\frac{3}{15}+\frac{5}{45}+\frac{7}{135}+\ldots\ldots\ldots\ldots+\frac{2n+1}{5(3)^n}$$

Subtracting:

$$\frac{2S_n}{3}=\frac{3}{5}+\frac{2}{15}+\frac{2}{45}+\frac{2}{135}+\ldots\ldots\ldots\ldots-\frac{2n+1}{5(3)^n}$$

$$\frac{2S_n}{3}=\frac{3}{5}+\frac{2}{15}\left(1+\frac{1}{3}+\frac{1}{9}+\ldots(n-1)\text{ terms}\right)-\frac{2n+1}{5(3)^n}$$

$$\frac{2S_n}{3}=\frac{3}{5}+\frac{2}{15}\left\{\frac{1-\frac{1}{3^{n-1}}}{1-\frac{1}{3}}\right\}-\frac{2n+1}{5(3)^n}$$

$$\frac{2S_n}{3}=\frac{3}{5}+\frac{1}{5}\left\{1-\frac{1}{3^{n-1}}\right\}-\frac{2n+1}{5(3)^n}$$

when $n\to\infty$ then $\frac{1}{3^{n-1}}=0$ and also $\frac{1}{5}\frac{2n+1}{3^n}=0$

hence $S_\infty=\frac{9}{10}+\frac{3}{10}=\frac{6}{5}$

Q.5 Find the sum to n terms and also find S_∞ for $\frac{3}{19}+\frac{33}{19^2}+\frac{333}{19^3}+\frac{3333}{19^4}+\ldots$

Sol. Let

$$S_n=3\left(\frac{1}{19}+\frac{11}{19^2}+\ldots n\text{ terms}\right)$$

$$S_n=\frac{3}{9}\left(\frac{9}{19}+\frac{99}{19^2}+\frac{999}{19^3}+\ldots n\text{ terms}\right)$$

$$S_n=\frac{1}{3}\left(\frac{10-1}{19}+\frac{10^2-1}{19^2}+\frac{10^3-1}{19^3}+\ldots+\frac{10^n-1}{19^n}\right)$$

$$S_n = \frac{1}{3}\left[\left\{\frac{10}{19}+\left(\frac{10}{19}\right)^2+\left(\frac{10}{19}\right)^3+\ldots+\left(\frac{10}{19}\right)^n\right\}-\left\{\frac{1}{19}+\frac{1}{19^2}+\frac{1}{19^3}+\ldots+\frac{1}{19^n}\right\}\right]$$

$$S_n = \frac{1}{3}\left[\frac{10}{19}\left(\frac{1-\left(\frac{10}{19}\right)^n}{1-\frac{10}{19}}\right)-\frac{1}{19}\left(\frac{1-\frac{1}{19^n}}{1-\frac{1}{19}}\right)\right]$$

$$S_n = \frac{1}{3}\left\{\frac{10}{9}\left(1-\left(\frac{10}{19}\right)^n\right)-\frac{1}{18}\left(1-\frac{1}{19^n}\right)\right\}$$

when $n\to\infty$ then $\left(\frac{10}{19}\right)^n = 0$ & $\frac{1}{19^n} = 0$

hence $S_\infty = \frac{1}{3}\left\{\frac{10}{9}-\frac{1}{18}\right\} = \frac{1}{3}\cdot\frac{19}{18} = \frac{19}{54}$

MISCELLANEOUS SEQUENCES

Type-1 :

Sequence dealing with $\Sigma n, \Sigma n^2, \Sigma n^3$

(1) $\Sigma n = \frac{n(n+1)}{2}$

(2) $\Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$

Proof :

consider the identity

$k^3 - (k-1)^3 = 3k^2 - 3k + 1$

for $k = 1 \Rightarrow 1^3 - 0^3 = 3(1^2) - 3(1) + 1$

$k = 2 \Rightarrow 2^3 - 1^3 = 3(2^2) - 3(2) + 1$

$k = 3 \Rightarrow 3^3 - 2^3 = 3(3^2) - 3(3) + 1$

:

:

$k = n \Rightarrow n^3 - (n-1)^3 = 3(n^2) - 3(n) + 1$

$$n^3 = 3\left(\Sigma n^2\right) - 3\left(\Sigma n\right) + n$$

$$\Rightarrow \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$$

(3) $\Sigma n^3 = \left[\frac{n(n+1)}{2}\right]^2 = (\Sigma n)^2$

Proof :

consider the identity

$k^4 - (k-1)^4 = 4k^3 - 6k^2 + 4k - 1$

for $k = 1 \Rightarrow 1^4 - 0^4 = 4(1^3) - 6(1^2) + 4(1) - 1$

$k = 2 \Rightarrow 2^4 - 1^4 = 4(2^3) - 6(2^2) + 4(2) - 1$

$k = 3 \Rightarrow 3^4 - 2^4 = 4(3^3) - 6(3^2) + 4\,(3) - 1$

$\vdots$

$K = n \Rightarrow n^4 - (n-1)^4 = 4(n^3) - 6(n^2) + 4(n) - 1$

$n^4 = 4(\Sigma n^3) - 6(\Sigma n^2) + 4(\Sigma n) - n$

$\Rightarrow \Sigma n^3 = \frac{n^2(n+1)^2}{4}$

Note :

(i) $\sum_{r=1}^{n}(a_r + b_r) = \sum_{r=1}^{n} a_r + \sum_{r=1}^{n} b_r$

(ii) $\sum_{r=1}^{n} k\, a_r = k\sum_{r=1}^{n} a_r$ (K $\in$ constant)

(iii) $\sum_{k=1}^{n} k = k\sum_{r=1}^{n} = k\,n$

Q.1 Evaluate : $\frac{1}{2}\left[\Sigma n^2 + \Sigma n\right]$

Sol. Given equation is

$$= \frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right]$$

$$= \frac{1}{2}\frac{n(n+1)}{2}\left[\frac{2n+1}{3} + 1\right]$$

$$= \frac{1}{4}n(n+1)\frac{2(n+2)}{3} = \frac{1}{6}n(n+1)(n+2)$$

Q.2 Find the value(s) of the positive integer n for which the quadratic equation, $\sum_{k=1}^{n}(x+k-1)(x+k)=10n$ has solutions α and $\alpha+1$ for some α.

Sol. Given equation is

$$\sum_{k=1}^{n}[x^2+(2k-1)x+k(k-1)]=10n$$

$$\Rightarrow x^2\sum_{k=1}^{n}1+x\sum_{k=1}^{n}(2k-1)+\sum_{k=1}^{n}(k^2-k)=10n$$

$$\Rightarrow nx^2+n^2x+\frac{n(n+1)(2n+1)}{6}-\frac{n(n+1)}{2}=10n$$

$$\Rightarrow nx^2+n^2x+\frac{n(n+1)}{2}\left(\frac{2n+1}{3}-1\right)=10n$$

$$\Rightarrow nx^2+n^2x+\frac{n(n+1)(n-1)}{3}=10n$$

$$\Rightarrow 3x^2+3nx+n^2-1=30$$

$\Rightarrow 3x^2 + 3nx + (n^2 - 31) = 0$ ($\alpha + 1$, α are its roots)

$\therefore$ difference of roots =1

$$\{(\alpha+1)-\alpha\}^2=1$$

$$\{(\alpha+1)+\alpha\}^2-4(\alpha+1)(\alpha)=1$$

$$\Rightarrow\left(-\frac{3n}{3}\right)^2-4\left(\frac{n^2-31}{3}\right)=1$$

$$\Rightarrow n^2-\frac{4}{3}n^2+\frac{124}{3}-1=0$$

$$\Rightarrow\frac{n^2}{3}=\frac{121}{3}\Rightarrow n=11$$

Q.2 Compute the sum : $(31)^2 + (32)^2 + (33)^2 + ... + (50)^2$

Sol. Required sum is

$$(1^2+2^2+3^2+\ldots+50^2)-(1^2+2^2+\ldots+30^2)$$

$$\Rightarrow\frac{50.51.101}{6}-\frac{30.31.61}{6}$$

$$= 33470$$

Q.4 Compute the sum of the series whose n^{th} term is given by

$T_n = n(n+1)(3n-1)$

Sol. Sum

$$S_n = \sum_{r=1}^{n} T_r = \sum_{r=1}^{n} r(r+1)(3r-1)$$

$$= \sum_{r=1}^{n} \left(3r^3 + 2r^2 - r\right)$$

$$= 3\sum_{r=1}^{n} r^3 + 2\sum_{r=1}^{n} r^2 - \sum_{r=1}^{n} r$$

$$= 3\frac{n^2(n+1)^2}{4} + \frac{2n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2}$$

$$\frac{3}{4}n^2(n+1)^2 + \frac{n}{3}(n+1)(2n+1) - \frac{n(n+1)}{2}$$

Q.5 $3^2 + 7^2 + 11^2 + \ldots$ (sum to n terms)

Sol. $T_r = (4r-1)^2$

$T_r = 16r^2 - 8r + 1$

$$S_n = \sum_{r=1}^{n} T_r = \sum_{r=1}^{n} 16r^2 - 8r + 1$$

$$S_n = 16\sum_{r=1}^{n} r^2 - 8\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$

$$= 16\frac{n(n+1)(2n+1)}{6} - 8\frac{n(n+1)}{2} + n = \frac{8}{3}n(n+1)(2n+1) - 4n(n+1) + n$$

Q.6 If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)\left(2^{n+1} - n - 2\right)$ where $n>1$, and the runs scored in the kth match are given by $k \cdot 2^{n+1-k}$, where $1 \le k \le n$, find n.

Sol. Given that

$$S_n = \left(\frac{n+1}{4}\right)\left(2^{n+1} - n - 2\right)$$

also $T_k = k.2^{n+1-k} = 2^{n+1}\left(\frac{k}{2^k}\right)$

now $S_n = \sum_{k=1}^{n} T_k$

$$S_n = 2^{n+1}\left(\frac{1}{2}\right) + 2^{n+1}\left(\frac{2}{2^2}\right) + \ldots + 2^{n+1}\left(\frac{n}{2^n}\right)$$

$$\frac{S_n}{2} = \qquad 2^{n+1}\left(\frac{1}{2^2}\right) + \ldots + 2^{n+1}\left(\frac{n-1}{2^n}\right) + 2^{n+1}\cdot\frac{n}{2^{n+1}}$$

$$\frac{S_n}{2} = 2^n + 2^{n-1} + \ldots + 2 - n$$

$$\frac{S_n}{2} = \frac{2\left(2^n - 1\right)}{(2-1)} - n$$

$$\Rightarrow S_n = 2^2\left(2^n - 1\right) - 2n$$

$$S_n = 2^{n+2} - 2n - 4$$

$$= 2\left(2^{n+1} - n - 2\right)$$

by comparison $\frac{n+1}{4} = 2 \Rightarrow n = 7$

Type-2 (Using method of difference) :

If T_1, T_2, T_3, ... are terms of a sequence then the terms $T_2 - T_1$, $T_3 - T_2$, $T_4 - T_3$, ... sometimes are in AP and sometimes in GP. For such series we first compute their n^{th} term and then compute the sum to n terms, using sigma notation.

Q.1 6 + 13 + 22 + 33 + ...n terms

Sol. Let

$$S = 6 + 13 + 22 + 33 + \ldots + T_n$$

$$S = \quad 6 + 13 + 22 + \ldots + T_{n-1} + T_n$$

$$0 = 6 + \underbrace{7 + 9 + 11 + \ldots}_{n-1 \text{ terms}} - T_n$$

$$T_n = 6 + \frac{(n-1)}{2}\left[2(7) + (n-2)2\right] = 6 + (n-1)(n+5)$$

$$T_n = n^2 + 4n + 1$$

$$S_n = \sum_{r=1}^{n} T_r = \frac{n(n+1)(2n+1)}{6} + \frac{4n(n+1)}{2} + n$$

$$= \frac{n(n+1)(2n+1)}{6} + 2n(n+1) + n$$

Q.2 3 + 8 + 15 + 24 + ... upto n terms.

Sol. Let

$$S = 3 + 8 + 15 + 24 + \ldots + T_n$$

$$S = \quad 3 + 8 + 15 + \ldots + T_{n-1} + T_n$$

$$0 = \underbrace{3 + 5 + 7 + 9 + \ldots}_{n \text{ terms}} - T_n$$

$$T_n = \frac{n}{2}\left[2(3) + (n-1)2\right] = n^2 + 2n$$

$$S_n = \sum_{r=1}^{n} T_r = \frac{n(n+1)(2n+1)}{6} + 2.\frac{n(n+1)}{2} = \frac{n(n+1)(2n+1)}{6} + n(n+1)$$

Q.3 5 + 7 + 13 + 31 + 85 + ...n terms

Sol. Let

$$S = 5 + 7 + 13 + 31 + 85 + \ldots + T_n$$

$$S = \quad 5 + 7 + 13 + 31 + \ldots + T_{n-1} + T_n$$

$$0 = 5 + \underbrace{2 + 6 + 18 + 54 + \ldots}_{n-1 \text{ terms}} - T_n$$

$$T_n = 5 + \frac{2\left(3^{n-1} - 1\right)}{(3-1)} = 3^{n-1} + 4$$

$$S_n = \sum_{r=1}^{n} T_r = \sum_{r=1}^{n} 3^{r-1} + \sum_{r=1}^{n} 4$$

$$S_n = \frac{1\left(3^n - 1\right)}{3-1} + 4n = \frac{1}{2}\left(3^n - 1\right) + 4n$$

Q.4 2 + 5 + 14 + 41 + 122 + n terms

Sol. Let

$$S = 2 + 5 + 14 + 41 + 122 + \ldots + T_n$$

$$S = \quad 2 + 5 + 14 + 41 + \ldots + T_{n-1} + T_n$$

$$0 = 2 + \underbrace{3 + 9 + 27 + 81 + \ldots}_{n-1 \text{ terms}} - T_n$$

$$T_n = 2 + \frac{3\left(3^{n-1} - 1\right)}{(3-1)} = 2 + \frac{3^n}{2} - \frac{3}{2} = \frac{3^n}{2} + \frac{1}{2}$$

$$S_n = \sum_{r=1}^{n} T_r = \frac{1}{2}\frac{3\left(3^n - 1\right)}{3-1} + \frac{n}{2} = \frac{3\left(3^n - 1\right)}{4} + \frac{n}{2}$$

Q.5 $1 + \left(1 + \frac{1}{3}\right) + \left(1 + \frac{1}{3} + \frac{1}{3^2}\right) + \ldots + \left(1 + \frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{n-1}}\right)$

Sol. For the above series

$$T_r = \left(1 + \frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{r-1}}\right)$$

$$T_r = \left(\frac{1 - \frac{1}{3^r}}{1 - \frac{1}{3}}\right) = \frac{3}{2}\left(1 - \frac{1}{3^r}\right) = \frac{3}{2} - \frac{1}{2}\left(\frac{1}{3^{r-1}}\right)$$

$$s_n = \sum_{r=1}^{n} T_r = \sum_{r=1}^{n}\left[\frac{3}{2} - \frac{1}{2}\left(\frac{1}{3^{r-1}}\right)\right]$$

$$s_n = \frac{3n}{2} - \frac{1}{2} \cdot \frac{1\left(1 - \frac{1}{3^n}\right)}{\left(1 - \frac{1}{3}\right)}$$

$$s_n = \frac{3n}{2} - \frac{3}{4}\left(1 - \frac{1}{3^n}\right)$$

Types-3 (Splitting the nth term as a difference of two):

(a) Here is a series in which each term is composed of the reciprocal of the product of r consecutive terms of AP.

Q. (i) $\frac{1}{1.2.3.4}+\frac{1}{2.3.4.5}+\frac{1}{3.4.5.6}+\ldots$

Sol. $T_r=\frac{1}{r(r+1)(r+2)(r+3)}=\frac{(r+3)-r}{3r(r+1)(r+2)(r+3)}$

$T_r=\frac{1}{3r(r+1)(r+2)}-\frac{1}{3(r+1)(r+2)(r+3)}$

now $T_1=\frac{1}{3(1.2.3)}-\frac{1}{3(2.3.4)}$

$T_2=\frac{1}{3(2.3.4)}-\frac{1}{3(3.4.5)}$

$T_3=\frac{1}{3(3.4.5)}-\frac{1}{3(4.5.6)}$

$\vdots$

$T_n=\frac{1}{3n(n+1)(n+2)}-\frac{1}{3(n+1)(n+2)(n+3)}$

on addition we get

$S_n=\frac{1}{3}\left\{\frac{1}{1.2.3}-\frac{1}{(n+1)(n+2)(n+3)}\right\}$

and $S_\infty=\frac{1}{3}\left\{\frac{1}{1.2.3}-0\right\}=\frac{1}{18}$

Q. (ii) $\frac{1}{1\cdot3\cdot5}+\frac{1}{3\cdot5\cdot7}+\frac{1}{5\cdot7\cdot9}+\ldots$

Sol. $T_r=\frac{1}{(2r-1)(2r+1)(2r+3)}$

$T_r=\frac{(2r+3)-(2r-1)}{4(2r-1)(2r+1)(2r+3)}$

$=\frac{1}{4(2r-1)(2r+1)}-\frac{1}{4(2r+1)(2r+3)}$

now

$T_1=\frac{1}{4(1.3)}-\frac{1}{4(3.5)}$

$T_2=\frac{1}{4(3.5)}-\frac{1}{4(5.7)}$

$$T_3 = \frac{1}{4(5.7)} - \frac{1}{4(7.9)}$$

$$\vdots$$

$$T_n = \frac{1}{4(2n-1)(2n+1)} - \frac{1}{4(2n+1)(2n+3)}$$

$$S_n = \frac{1}{4}\left\{\frac{1}{3} - \frac{1}{(2n+1)(2n+3)}\right\}$$

and $S_\infty = \frac{1}{4}\left\{\frac{1}{3} - 0\right\} = \frac{1}{12}$

(b) In this case a factor is missing

Q. **(iii)** $\frac{3}{1.2.4} + \frac{4}{2.3.5} + \frac{5}{3.4.6} + \ldots$ then split the nth term and solve

Sol. $$T_r = \frac{(r+2)}{r(r+1)(r+3)} = \frac{(r+2)^2}{r(r+1)(r+2)(r+3)}$$

$$= \frac{(r+1)(r+3)+1}{r(r+1)(r+2)(r+3)}$$

$$= \frac{1}{r(r+2)} + \frac{1}{r(r+1)(r+2)(r+3)}$$

$$= \frac{(r+2)-r}{2r(r+2)} + \frac{(r+3)-r}{3r(r+1)(r+2)(r+3)}$$

$$= \frac{1}{2r} - \frac{1}{2(r+2)} + \frac{1}{3r(r+1)(r+2)} - \frac{1}{3(r+1)(r+2)(r+3)}$$

$$T_1 = \frac{1}{2(1)} - \frac{1}{2(3)} + \frac{1}{3(1.2.3)} - \frac{1}{3(2.3.4)}$$

$$T_2 = \frac{1}{2(2)} - \frac{1}{2(4)} + \frac{1}{3(2.3.4)} - \frac{1}{3(3.4.5)}$$

$$T_3 = \frac{1}{2(3)} - \frac{1}{2(5)} + \frac{1}{3(3.4.5)} - \frac{1}{3(4.5.6)}$$

$$\vdots$$

$$T_{n-1} = \frac{1}{2(n-1)} - \frac{1}{2(n+1)} + \frac{1}{3(n-1)n(n+1)} - \frac{1}{3n(n+1)(n+2)}$$

$$T_n = \frac{1}{2(n)} - \frac{1}{2(n+2)} + \frac{1}{3n(n+1)(n+2)} - \frac{1}{3(n+1)(n+2)(n+3)}$$

$$S_n = \frac{1}{2}\left\{1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right\} + \frac{1}{3}\left\{\frac{1}{6} - \frac{1}{(n+1)(n+2)(n+3)}\right\}$$

Now $S_\infty = \frac{1}{2}\left(1 + \frac{1}{2}\right) + \frac{1}{3}\left(\frac{1}{6}\right) = \frac{3}{4} + \frac{1}{18} = \frac{29}{36}$

(c) Thinking Based

Q. **(iv)** $\frac{1}{1.3} + \frac{2}{1.3.5} + \frac{3}{1.3.5.7} + \frac{4}{1.3.5.7.9} + \ldots$

Sol. $T_r = \frac{r}{1.3.5.7\ldots(2r+1)}$

$$T_r = \frac{1}{2}\left\{\frac{(2r+1)-1}{1.3.5.7\ldots(2r+1)}\right\}$$

$$T_r = \frac{1}{2}\left\{\frac{1}{1.3.5\ldots.(2r-1)} - \frac{1}{3.5.7.(2r+1)}\right\}$$

$$T_1 = \frac{1}{2}\left\{\frac{1}{1} - \frac{1}{3}\right\}$$

$$T_2 = \frac{1}{2}\left\{\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5}\right\}$$

$$T_3 = \frac{1}{2}\left\{\frac{1}{1\cdot 3\cdot 5} - \frac{1}{3\cdot 5\cdot 7}\right\}$$

$$\vdots$$

$$\vdots$$

$$T_n = \frac{1}{2}\left\{\frac{1}{1\cdot 3\cdot 5\ldots(2n-1)} - \frac{1}{3\cdot 5\cdot 7\ldots(2n+1)}\right\}$$

$$s_n = \frac{1}{2}\left\{1 - \frac{1}{3.5.7.(2n+1)}\right\}$$

also $S_\infty = \frac{1}{2}\{1-0\} = \frac{1}{2}$

Q. **(v)** Find S_n for $\frac{1}{2.4}+\frac{1.3}{2.4.6}+\frac{1.3.5}{2.4.6.8}+\ldots$

Sol. $T_r=\frac{1.3.5\ldots(2r-1)}{2.4.6\ldots(2r+2)}$

$$T_r=\frac{1.3.5\ldots(2r-1)\{(2r+2)-(2r+1)\}}{2.4.6\ldots(2r+2)}$$

$$T_r=\frac{1.3.5\ldots(2r-1)}{2.4.6\ldots(2r)}-\frac{1.3.5\ldots(2r+1)}{2.4.6\ldots.(2r+2)}$$

Now

$$T_1=\frac{1}{2}-\frac{1\cdot3}{2\cdot4}$$

$$T_2=\frac{1\cdot3}{2\cdot4}-\frac{1\cdot3\cdot5}{2\cdot4\cdot6}$$

$$T_3=\frac{1\cdot3\cdot5}{2\cdot4\cdot6}-\frac{1\cdot3\cdot5\cdot7}{2\cdot4\cdot6\cdot8}$$

$$\vdots$$

$$T_n=\frac{1\cdot3\cdot5\ldots(2n-1)}{2\cdot4\cdot6\ldots(2n)}-\frac{1\cdot3\cdot5\ldots(2n+1)}{2\cdot4\cdot6\ldots(2n+2)}$$

$$S_n=\frac{1}{2}-\frac{1.3.5\ldots(2n+1)}{2.4.6\ldots(2n+2)}$$

Q.1 Find the sum of the n terms of the sequence

$$\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\ldots$$

Sol. $T_r=\frac{r}{1+r^2+r^4}=\frac{r}{(1+r+r^2)(1-r+r^2)}$

$$T_r=\frac{(r^2+r+1)-(r^2-r+1)}{2(1+r+r^2)(1-r+r^2)}$$

$$T_r=\frac{1}{2(1-r+r^2)}-\frac{1}{2(1+r+r^2)}$$

$$T_1=\frac{1}{2(1)}-\frac{1}{2(3)}$$

$$T_2 = \frac{1}{2(3)} - \frac{1}{2(7)}$$

$$T_3 = \frac{1}{2(7)} - \frac{1}{2(13)}$$

:

$$T_n = \frac{1}{2(1-n+n^2)} - \frac{1}{2(1+n+n^2)}$$

$$S_n = \frac{1}{2}\left\{1 - \frac{1}{1+n+n^2}\right\}$$

Also, $S_\infty = \frac{1}{2}\{1-0\} = \frac{1}{2}$

Q.2 Let f(x) denote the sum of the infinite trigonometric series, $f(x) = \sum_{n=1}^{\infty} \sin\frac{2x}{3^n}.\sin\frac{x}{3^n}$.

Find f(x) (independent of n). If the sum of the solution of the equation f(x) = 0 lie in in the interval (0, 629) is $2k\pi$, find k

Sol. Let $\frac{x}{3^n} = \theta$

then $\sin2\theta.\sin\theta = 2\sin^2\theta\cos\theta$

$$= 2\left(1-\cos^2\theta\right)\cos\theta$$

$$= 2\cos\theta - 2\cos^3\theta$$

$$= \frac{1}{2}\left[4\cos\theta - 4\cos^3\theta\right]$$

$$= \frac{1}{2}\left[\cos\theta - \left(4\cos^3\theta - 3\cos\theta\right)\right]$$

$$= \frac{1}{2}\left[\cos\theta - \cos3\theta\right]$$

hence $T_n = \sin\frac{2x}{3^n}.\sin\frac{x}{3^n} = \frac{1}{2}\cos\frac{x}{3^n} - \frac{1}{2}\cos\frac{x}{3^{n-1}}$

Now

$$T_1 = \frac{1}{2}\cos\frac{x}{3} - \frac{1}{2}\cos x$$

$$T_2 = \frac{1}{2}\cos\frac{x}{3^2} - \frac{1}{2}\cos\frac{x}{3}$$

$$T_3 = \frac{1}{2}\cos\frac{x}{3^3} - \frac{1}{2}\cos\frac{x}{3^2}$$

$$\vdots$$

$$\vdots$$

$$T_n = \frac{1}{2}\cos\frac{x}{3^n} - \frac{1}{2}\cos\frac{x}{3^{n-1}}$$

$$S_n = \frac{1}{2}\left(\cos\frac{x}{3^n} - \cos x\right)$$

$$\Rightarrow S_\infty = \frac{1}{2}(\cos 0 - \cos x) = \frac{1}{2}(1 - \cos x) = f(x)$$

now equation $f(x) = 0 \Rightarrow \cos x = 1$

$\Rightarrow x = 2\pi, 4\pi, \ldots 200\pi$

Sum of all solutions $= 2\pi(1 + 2 + \ldots + 100)$

$$= 2\pi\frac{100.101}{2} \Rightarrow k = 5050$$

Type-4

Here is a series in which each term is composed of r factors in AP. the first factor of the several terms being in the same AP.

Q. 1.2.3.4 + 2.3.4.5 + 3.4.5.6 +... upto n terms

Sol. $T_r = r(r+1)(r+2)(r+3)$

$$T_r = \frac{1}{5}r(r+1)(r+2)(r+3)\{(r+4) - (r-1)\}$$

$$T_r = \frac{1}{5}r(r+1)(r+2)(r+3)(r+4) - \frac{1}{5}(r-1)r(r+1)(r+2)(r+3)$$

$$T_1 = \frac{1}{5}(1\cdot 2\cdot 3\cdot 4\cdot 5) - \frac{1}{5}(0\cdot 1\cdot 2\cdot 3\cdot 4)$$

$$T_2 = \frac{1}{5}(2\cdot 3\cdot 4\cdot 5\cdot 6) - \frac{1}{5}(1\cdot 2\cdot 3\cdot 4\cdot 5)$$

$$T_3 = \frac{1}{5}(3\cdot 4\cdot 5\cdot 6\cdot 7) - \frac{1}{5}(2\cdot 3\cdot 4\cdot 5\cdot 6)$$

$$\vdots$$

$$T_n = \frac{1}{5}n(n+1)(n+2)(n+3)(n+4) - \frac{1}{5}(n-1)n(n+1)(n+2)(n+3)$$

$$S_n = \frac{1}{5}n(n+1)(n+2)(n+3)(n+4)$$

Harmonic Progression (HP):

A standard HP is $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \ldots, \frac{1}{a+\overline{n-1}d}$

Definition

A sequence is said to be in HP if the reciprocal of its terms are in AP.

Note :

(i) If the term of a HP is not defined this means that the corresponding term of the AP is zero.

(ii) There is no general formula for finding the sum to n terms of HP.

(iii) If a, b, c are in HP

$$b = \frac{2ac}{a+c} \text{ or } \frac{a}{c} = \frac{a-b}{b-c}$$

Q.1 If the 3rd , 6th and last term of a HP are $\frac{1}{3}, \frac{1}{5}, \frac{3}{203}$ respectively, find the number of terms.

Sol. Given $T_3 = \frac{1}{3} \Rightarrow \frac{1}{a+2d} = \frac{1}{3} \Rightarrow a + 2d = 3$ ...(1)

$T_6 = \frac{1}{5} \Rightarrow \frac{1}{a+5d} = \frac{1}{5} \Rightarrow a + 5d = 5$ ...(2)

also $T_n = \frac{3}{203} \Rightarrow \frac{1}{a+(n-1)d} = \frac{3}{203} \Rightarrow a + (n-1)d = \frac{203}{3}$ ...(3)

(2) – (1) $\Rightarrow 3d = 2 \Rightarrow d = 2/3$

and $a = 3 - 2d = 3 - \frac{4}{3} = \frac{5}{3}$

from (3), $\frac{5}{3} + (n-1)\frac{2}{3} = \frac{203}{3} \Rightarrow (n-1)2 = 198 \Rightarrow n = 100$

Q.2 If m^{th} term of a HP is n, and n^{th} term is equal to m then prove that $(m+n)^{th}$ term is $\frac{mn}{m+n}$.

Sol. Given $T_m = n \Rightarrow a + (m-1)d = \frac{1}{n}$ ...(1)

also $T_n = m \Rightarrow a + (n-1)d = \frac{1}{m}$ ...(2)

(2) – (1)

$$(n-m)d = \frac{n-m}{mn} \Rightarrow d = \frac{1}{mn}$$

from (1), $\Rightarrow a + (m-1)\frac{1}{nm} \Rightarrow a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{n} \Rightarrow a = \frac{1}{mn}$

now $T_{m+n} = \frac{1}{a+(m+n-1)d} = \frac{mn}{1+(m+n-1)} = \frac{mn}{m+n}$

Q.3 If $a_1, a_2, a_3, \ldots a_n$ are in HP then prove that

$$a_1a_2 + a_2a_3 + a_3a_4 + \ldots + a_{n-1}a_n = (n-1)a_1a_n$$

Sol. Clearly

$$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots, \frac{1}{a_n} \text{ are in AP}$$

now $d = \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} = \ldots = \frac{1}{a_n} - \frac{1}{a_{n-1}}$

$$d = \frac{a_1 - a_2}{a_1a_2} = \frac{a_2 - a_3}{a_2a_3} = \ldots = \frac{a_{n-1} - a_n}{a_{n-1}a_n}$$

now $a_1a_2 + a_2a_3 + \ldots + a_{n-1}a_n = \frac{a_1 - a_2}{d} + \frac{a_2 - a_3}{d} + \ldots + \frac{a_{n-1} - a_n}{d}$

$$= \frac{a_1 - a_n}{d} \qquad \ldots(1)$$

Also we know $\frac{1}{a_n} = \frac{1}{a_1} + (n-1)d$

$$\Rightarrow \frac{a_1 - a_n}{a_1a_n} = (n-1)d \Rightarrow \frac{a_1 - a_n}{d} = (n-1)a_1a_n \qquad \ldots(2)$$

from (1) and (2)

$$a_1a_2 + a_2a_3 + \ldots + a_{n-1}a_n = (n-1)a_1a_n$$ **Hence proved.**

Q.4 If a, b, c are in H. P, find the value of $\frac{b+a}{b-a} + \frac{b+c}{b-c}$

Sol. $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ in AP

$$\frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b} = d(\text{let})$$

$$\frac{a-b}{ab} = \frac{b-c}{bc} = d$$

now given exp. $\frac{b+a}{-abd}+\frac{b+c}{bcd}$

$$=\frac{1}{bd}\left\{\frac{-b}{a}-1+\frac{b}{c}+1\right\}\Rightarrow\frac{1}{bd}b\left(\frac{1}{c}-\frac{1}{a}\right)=\frac{2d}{d}=2$$

HARMONIC MEAN

Consider n positive quantities $a_1, a_2, a_3, ..., a_n$.

$$H.M.=\frac{n}{\frac{1}{a_1}+\frac{1}{a_2}+...+\frac{1}{a_n}}$$

Note :

If a, b, c are in HP then middle term is called the harmonic mean between them. Hence if H is the harmonic mean (H.M.) of a and b then a, H, b are in HP and $H=\frac{2ab}{a+b}$.

To insert n HM between a and b

Let $H_1, H_2, ...H_n$ are n HM's between a and b

hence $\sum_{i=1}^{n}\frac{1}{H_i}=n.\frac{1}{H}$

Hence sum of the reciprocals of all n HM's between a and b is equal to n times a single HM between a and b.

Q.1 Find the HM between the roots of the equation $x^2 - 10x + 11 = 0$

Sol. Let roots are α, β then $\alpha + \beta = 10$, $\alpha\beta = 11$

$$\text{HM of } \alpha,\beta=\frac{2}{\frac{1}{\alpha}+\frac{1}{\beta}}=\frac{2(11)}{10}=\frac{11}{5}$$

Q.2 Between 1 and $\frac{1}{100}$ if 100 HM are inserted then $\sum_{i=1}^{100}\frac{1}{H_i}=?$

Sol. HM of a and b $(a=1, b=\frac{1}{100})$

$$H = \frac{2ab}{a+b} = \frac{2(1)\left(\frac{1}{100}\right)}{1+\frac{1}{100}} = \frac{2}{101}$$

$$\text{also } \sum_{i=1}^{100} \frac{1}{H_i} = 100.\frac{1}{H} = 100.\frac{101}{2} = 5050$$

Q.3 If a is the AM of b and c, b is GM of c and a then prove that c is the HM of a and b.

Sol. Given $a = \frac{b+c}{2} \Rightarrow 2a - b = c$ ...(1)

also $b^2 = ac \Rightarrow b^2 = a(2a-b)$

$$b^2 - a^2 = a(a-b)$$

$$(b-a)(b+a) = a(a-b) \Rightarrow a = b \text{ or } b + a = -a$$

$$b = -2a$$

If a = b then $c = 2a - a = a$ from (1)

$\therefore$ a = b = c $\Rightarrow$ c is HM of a & b

If b = – 2a then c = 2a – (–2a) = 4a

Also $\frac{2ab}{a+b} = \frac{2a(-2a)}{a+(-2a)} = 4a = c \Rightarrow$ c is HM of a and b.

Q.4 If a^2, b^2, c^2 are in AP, show that b + c, c + a, a + b are in HP

Sol. $\because$ a^2, b^2, c^2 in AP

add ab + bc + ca

$\Rightarrow$ (a + b)(c + a), (a + b)(b + c), (b + c)(c + a)

Divide by (a + b)(b + c)(c + a)

$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ in AP

$\Rightarrow$ b + c, c + a, a + b in HP

Q.5 The values of xyz is $\frac{15}{2}$ or $\frac{18}{5}$ according as the series a, x, y, z, b is an AP or HP. Find the values of a & b assuming them to be positive integer.

Sol. Given

If a, x, y, z, b in AP $\Rightarrow$ $xyz = \frac{15}{2}$

now
b, z, y, x, a also AP
divide by ab

$\frac{1}{a}, \frac{z}{ab}, \frac{y}{ab}, \frac{x}{ab}, \frac{1}{b}$ in AP

$\Rightarrow a, \frac{ab}{z}, \frac{ab}{y}, \frac{ab}{x}, b$ in HP

now given $\frac{ab}{z}.\frac{ab}{y}.\frac{ab}{x} = \frac{18}{5}$

$\Rightarrow (ab)^3 = \frac{18}{5}xyz = \frac{18}{5}.\frac{15}{2} = 27$

$\Rightarrow (ab)^3 = 3^3 \Rightarrow ab = 3$

hence a = 3, b = 1 or a = 1, b = 3

Q.6 If the roots of $10x^3 - cx^2 - 54x - 27 = 0$ are in harmonic progression, then find c and all the roots.

Sol. Let roots are α, β, γ (in HP)

the equation having roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ in AP is $\frac{10}{x^3} - \frac{c}{x^2} - \frac{54}{x} - 27 = 0$

$$\Rightarrow 27x^3 + 54x^2 + cx - 10 = 0 \begin{cases} \frac{1}{\alpha} = a - d \\ \frac{1}{\beta} = a \\ \frac{1}{\gamma} = a + d \end{cases}$$

sum = $3a = \frac{-54}{27} = -2 \Rightarrow a = \frac{-2}{3}$

product = $a\left(a^2 - d^2\right) = \frac{10}{27}$

$\Rightarrow \frac{-2}{3}\left(\frac{4}{9} - d^2\right) = \frac{10}{27} \Rightarrow \frac{4}{9} - d^2 = -\frac{5}{9}$

$\Rightarrow d^2 = 1 \Rightarrow d = \pm 1$

roots $\frac{1}{\alpha} = \frac{-2}{3} \pm 1, \frac{1}{\beta} = \frac{-2}{3}, \frac{1}{\gamma} = \frac{-2}{3} \mp 1$

$$(\alpha, \beta, \gamma) \equiv \left(3, \frac{-3}{2}, \frac{-3}{5}\right) \text{or} \left(\frac{-3}{5}, \frac{-3}{2}, 3\right)$$

Also, 3 is a root of given equation.
hence $10(27) - c(9) - 54(3) - 27 = 0$

$9\{30 - c - 18 - 3\} = 0 \Rightarrow c = 9$

Q.7 If the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal then prove that a, b, c are in H.P.

Sol. Clearly $x = 1$ is a root hence both roots are 1, 1

Now product $1.1 = \frac{c(a-b)}{a(b-c)}$

$\Rightarrow ca - cb = ab - ac$

$\Rightarrow 2ac = ab + bc$

$\Rightarrow b = \frac{2ac}{a+c}$

$\Rightarrow$ a, b, c are in HP.

Relation between AM, GM and HM

If a and b are two positive numbers then $A \geq G \geq H$ and A, G, H are in G. P i.e., $G^2 = AH$.

In fact, RMS $\geq$ AM $\geq$ GM $\geq$ HM

Q.1 If a, b and c are positive then prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$.

Sol. We have

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

$$\Rightarrow \frac{a}{b+c} + 1 + \frac{b}{c+a} + 1 + \frac{c}{a+b} + 1 \geq \frac{3}{2} + 3$$

$$\Rightarrow \frac{a+b+c}{b+c} + \frac{a+b+c}{c+a} + \frac{a+b+c}{a+b} \geq \frac{9}{2}$$

$$\Rightarrow \frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \geq \frac{9}{2(a+b+c)}$$

Now, using A.M. $\geq$ H.M., we have

$$\frac{\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}}{3} \geq \frac{3}{(a+b)+(b+c)+(c+a)}$$

$$\Rightarrow \quad \frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \geq \frac{9}{2(a+b+c)}$$

GENERAL ILLUSTRATIONS :

Q.1 If $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}$ and p, q, r are in AP then prove that x, y, z are in HP.

Sol. $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}=k\,(\text{let})$

$\frac{a-x}{kx}=p,\ \frac{a-y}{ky}=q,\ \frac{a-z}{kz}=r$

now p, q, r are in AP

hence $2q=p+r$

$$2\left(\frac{a-y}{ky}\right)=\frac{a-x}{kx}+\frac{a-z}{kz}$$

$$\Rightarrow\quad \frac{2}{k}\left(\frac{a}{y}-1\right)=\frac{1}{k}\left\{\frac{a}{x}-1+\frac{a}{z}-1\right\}$$

$$\Rightarrow\quad \frac{2}{y}=\frac{1}{x}+\frac{1}{z}\Rightarrow x,y,z \text{ in H.P}$$

Q.2 If $a^x=b^y=c^z=d^w$ and a, b, c, d are in GP then prove that x, y, z, w are in HP.

Sol. Let $a^x=b^y=c^z=d^w=k\,(\text{let})$

$$\Rightarrow\quad a=k^{\frac{1}{x}},b=k^{\frac{1}{y}},c=k^{\frac{1}{z}},d=k^{\frac{1}{w}}$$

$$\because\quad k^{\frac{1}{x}},k^{\frac{1}{y}},k^{\frac{1}{z}},k^{\frac{1}{w}} \text{ in GP}$$

$$\therefore\quad \frac{1}{x},\frac{1}{y},\frac{1}{z},\frac{1}{w} \text{ in AP}$$

$$\Rightarrow\quad x, y, z, w \text{ are in HP}$$

Q.3 If a, b, c are three distinct positive reals in HP then prove that $a^n+c^n>2b^n$

Sol. Since a, b, c in HP

therefore, b is HM of a and c

Also, GM > HM

$$\sqrt{ac}>b\Rightarrow\left(\sqrt{ac}\right)^n>b^n \qquad ...(1)$$

also let numbers a^n, c^n

AM > GM

$$\frac{a^n+c^n}{2}>\left(a^nc^n\right)^{\frac{1}{2}}\Rightarrow\frac{a^n+c^n}{2}>\left(\sqrt{ac}\right)^n \qquad ...(2)$$

by (1) and (2)

$$\frac{a^n + c^n}{2} > b^n \Rightarrow a^n + c^n > 2b^n$$

Q.4 If a, b, c are in AP. p, q, r are in HP and ap, bq, cr are in GP then prove that $\frac{p}{r} + \frac{r}{p} = \frac{a}{c} + \frac{c}{a}$.

Sol. a, b, c are in AP $\Rightarrow$ $2b = a + c$ ...(1)

p, q, r are in HP $\Rightarrow$ $q = \frac{2pr}{p+r}$ ...(2)

Also, ap, bq, cr are in GP

$\Rightarrow$ $b^2q^2 = ap.cr$

$\Rightarrow$ $\left(\frac{a+c}{2}\right)^2\left(\frac{2pr}{p+r}\right)^2 = ac.pr$

$\Rightarrow$ $\frac{a}{c} + \frac{c}{a} = \frac{p}{r} + \frac{r}{p}$

Q.5 If the roots of equation $x^3 - 11x^2 + 36x - 36 = 0$ are in H. P. find the middle root.

Sol. Let roots are α , β, γ (in HP)

then equation having roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ is $\frac{1}{x^3} - \frac{11}{x^2} + \frac{36}{x} - 36 = 0$

$36x^3 - 36x^2 + 11x - 1 = 0$ $\Leftarrow$ $\frac{1}{\alpha} = a - d$, $\frac{1}{\beta} = a$, $\frac{1}{\gamma} = a + d$

sum $= 3a = \frac{36}{36} = 1 \Rightarrow a = \frac{1}{3} = \frac{1}{\beta}$

$\therefore$ middle root $\beta = 3$